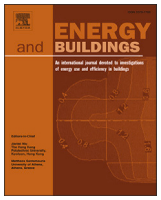
Energy & Buildings 325 (2024) 114993



Contents lists available at ScienceDirect

Energy & Buildings

journal homepage: www.elsevier.com/locate/enbuild





A stochastic MPC-based energy management system for integrating solar PV, battery storage, and EV charging in residential complexes

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ARTICLE INFO

Dataset link: https://drive.google.com/drive/folders/1XMkrhZrh-8I15SrLw9JxULVLej0HUUQj?usp=sharing

Keywords:
Solar photovoltaics
Energy management system
Battery energy storage system
Model predictive control
Electric vehicle charging

ABSTRACT

This paper presents a Stochastic Model Predictive Control (SMPC)-based energy management system (EMS) for residential complexes with integrated solar photovoltaics (PV), battery energy storage systems (BESS), and electric vehicle (EV) charging infrastructure. The EMS coordinates BESS operations, integrating solar generation, residential load demand, and EV charging. It optimizes BESS charging/discharging based on solar power, load demands, electricity pricing, and feed-in tariffs over a finite horizon, while considering uncertainties through multiple scenarios of load and EV charging demand, as well as solar generation. By accounting for battery degradation, cost savings, and revenue from energy transactions, the proposed EMS enhances BESS longevity and profitability. The EMS also manages reactive power provision from the BESS inverter, ensuring voltage stability in the presence of uncertainties. Extensive case studies on Matlab Simscape Electrical and real-time validation on the OPAL-RT simulator demonstrate the effectiveness of the proposed SMPC-based EMS in optimizing energy use, operational efficiency, and economic returns, contributing significantly to the sustainable energy management of the residential complex.

1. Introduction

1.1. Background and motivation

The transformation from traditional, fossil fuel-driven synchronous generators (SGs) to inverter-based renewable energy sources (RESs) is driven by diminishing fossil fuel reserves and an increasing commitment to sustainable energy practices [1–3]. As various countries transition from SGs to RESs [4,5], these changes align with efforts to meet both national and global greenhouse gas (GHG) reduction targets. This shift has led to the progressive replacement of conventional SGs with inverter-based RESs [6,7]. Additionally, the rising cost of grid electricity, coupled with decreasing costs of solar energy technologies and supportive policies, has driven substantial investments in rooftop solar photovoltaics (PV). Australia has seen a remarkable growth in installed capacity of rooftop solar PV over the past decade; in 2023, rooftop solar became the country's second largest source of renewable electricity, providing 11.2% of the nation's power and surpassing 20 GW of installed capacity [8]. Residential complexes, which are multi-unit housing developments that include apartments, town-houses, and other residential buildings, have been significant contributors to this growth. The adaptation of rooftop solar PV in these complexes is driven by both environmental and economic incentives, offering residents a way to reduce their carbon footprint and lower their electricity bills.

However, the inherent variability of solar PV, influenced by factors like solar irradiance and ambient temperature, poses challenges for grid stability as solar capacity expands [9–11]. Moreover, the residential electricity demand profile often does not align with solar generation due to variations in household energy consumption throughout the day and across seasons, as well as fluctuations in solar irradiance and weather conditions. Household electricity demand typically peaks in the morning and evening when solar generation is lower or non-existent, leading to reliance on grid electricity during these times. Conversely, solar generation peaks at midday when many households have lower electricity consumption, resulting in excess solar power that may not be fully utilized on-site and is often sold back to the grid at feed-in tariff rates that are much cheaper than grid electricity prices. Additionally, the rising popularity of electric vehicles (EVs) and the expansion of EV charging infrastructure add further complexity to managing electricity demand in residential complexes [12]. To best utilize the harvested energy from rooftop solar PV systems in residential complexes by aligning it with the electricity demand, it is necessary to implement effective energy management strategies. These include energy storage systems, demand

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https://doi.org/10.1016/j.enbuild.2024.114993
Received 22 July 2024; Received in revised form 21 October 2024; Accepted 1 November 2024
Available online 13 November 2024
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response programs, and smart grid technologies, which optimize solar energy use, reduce reliance on grid electricity, and enhance overall system efficiency and sustainability [13]. Battery energy storage systems (BESSs) are becoming increasingly popular in residential complexes. They maximize the benefits of harvested solar energy by storing excess solar power generated during periods of low demand for use during peak hours, aligning solar generation more closely with household electricity demand and reducing reliance on grid electricity. Additionally, BESSs help manage the increased electricity demand from the integration of EV charging infrastructure, ensuring efficient energy use and further reducing grid dependency [14,15].

Effective management of BESS is crucial to maximizing the economic benefits of residential complexes by optimizing solar energy utilization, reducing battery degradation, and efficiently managing residential and EV loads. Additionally, the increased penetration of solar PV can lead to voltage fluctuation issues due to the intermittent and variable nature of solar power generation, highlighting the importance of energy management strategies in residential complexes to address these fluctuations [16]. An efficient energy management strategy is pivotal for managing the load demand of the residential complex and EV charging demand. It leverages harvested solar energy, coordinates the charging and discharging of the BESS, exchanges power with the utility grid based on the tariff structure, and offers reactive power support to stabilize voltage levels at the point of common coupling (PCC), ensuring the economic and reliable operation of the residential complex.

1.2. Relevant literature

Research on EMS for residential complexes focuses on managing the interactions among RESs, BESSs, and EVs to optimize energy flows and enhance economic sustainability. However, significant gaps remain in addressing PCC voltage stability, BESS degradation, uncertainty management, and stochastic EV charging behavior.

Many EMS frameworks emphasize economic optimization but neglect essential aspects like voltage regulation, BESS degradation, and uncertainty management. For instance, [17] presents a day-ahead scheduling model minimizing costs under uncertain solar generation and load demand. Similarly, [18] explores cost savings by incorporating V2G operations. However, both models overlook BESS degradation, which could compromise long-term performance. Strategies using genetic algorithms, such as [19], optimize energy flows without accounting for battery wear, resulting in increased operational costs. Short-term cost-saving models, such as [20] and [21], regulate PV and BESS operations but lack degradation-aware strategies, limiting their sustainability. Multi-objective EMS frameworks like [22] and [23] aim to minimize costs and emissions but fail to address battery degradation impacts effectively. Although [24] integrates PV and wind forecasting, it does not fully account for BESS wear during high cycling demand, which can impair long-term performance. Computational efficiency is improved in [25], but this framework also neglects the cumulative effects of battery degradation.

Maintaining voltage stability at the PCC is critical, especially in weak-grid environments with high renewable penetration. Variability in solar generation and household demand can disrupt grid stability, leading to solar curtailment, load shedding, or operational issues if not effectively managed. Many EMS frameworks attempt to stabilize voltage by adjusting real power flows through BESS operations [26,27]. However, excessive reliance on real power regulation increases stress on BESSs, accelerating degradation. Since BESS operation is constrained by SoC limits and charging/discharging thresholds, real power adjustments alone are often insufficient to maintain stable voltage. Without reactive power support, rapid BESS cycling during solar peaks and peak load periods can cause voltage violations, further straining the system. Although reactive power support from BESS inverters offers a sustainable solution, many EMS frameworks do not incorporate it. For example, [28] applies particle swarm optimization for real-time management but omits reactive power support, leaving the system vulnerable to voltage instability. Similarly, [29] focuses on cost optimization without addressing reactive power needs. Advanced optimization techniques in [30] manage energy flows but neglect reactive power support, exposing the system to voltage fluctuations. In contrast, [31] balances voltage management and degradation considerations but relies heavily on real power adjustments, leading to frequent BESS cycling and accelerated wear. Similar approaches are reported in [32,33]. Although some EMS models incorporate BESS degradation management, they often neglect PCC voltage regulation. For instance, [34] integrates degradation-aware dispatch without focusing on PCC voltage management, potentially compromising system stability. Similarly, [35] models both cycle and calendar degradation but does not address PCC voltage control, leading to challenges under variable conditions. In [36], several EMS strategies are compared, revealing that battery cycling behavior can vary by up to 55% across strategies, impacting degradation. While the study emphasizes the need for managing battery wear, it does not address PCC voltage management.

A critical limitation of EMS frameworks is the lack of comprehensive uncertainty management. The inherent variability in solar generation, load demand, and EV charging behavior introduces unpredictability that many models fail to address adequately. Although studies such as [37], [38], and [39] adopt scenario-based approaches to manage uncertainties, they do not fully account for the combined impacts of solar variability, stochastic EV demand, and battery degradation. When EV charging demand is included, it is often treated as a simplified static load [40]. However, real-world EV usage introduces stochastic elements, including user preferences, charging habits, and time-of-use behavior. Proper management of these behaviors is essential to prevent system overload and enhance energy utilization.

1.3. Shortcomings in existing literature and novel features of the proposed approach

Based on the insights from the literature review, several limitations in existing EMS frameworks for residential complexes become evident:

- *Neglect of voltage support at the PCC:* As discussed in the previous section, ensuring voltage stability at the PCC is critical, particularly in weak grids with high renewable energy penetration, but it has not been adequately explored by existing EMSs. While some studies rely on real power adjustments for voltage regulation, this approach can accelerate BESS cycling and degradation. A few studies, such as [31], have addressed voltage management, but they primarily focus on real power adjustments, which are limited by SoC constraints and power thresholds.
- *Failure to adequately consider BESS degradation:* Existing EMSs mostly lack degradation-aware strategies for BESS operations. Frequent cycling without proper degradation management shortens battery lifespan and increases operational costs. Although some studies (e.g., [24], [36], [34]) incorporate BESS degradation into their models, they often overlook voltage regulation, limiting their effectiveness in ensuring long-term system sustainability.
- *Insufficient Uncertainty Management:* Many EMS frameworks inadequately address uncertainties related to load demand, solar generation, and EV charging behavior. Although some studies adopt scenario-based planning to manage uncertainties (e.g., [37], [38], [39]), they fail to fully account for the combined effects of these uncertainties on BESS operations and grid interactions.
- *Simplistic Treatment of EV Charging Demand:* Managing EV charging behavior effectively requires addressing its inherent uncertainty, alongside other stochastic elements such as solar generation and load fluctuations. However, many models treat EV demand as a static or predictable load [40], neglecting real-world stochastic elements such as user preferences and charging habits. Even frameworks that consider EV demand (e.g., [24]) often overlook the

Table 1
Comparison of attributes across various EMS frameworks in existing literature.

Ref.	Economics	Voltage Support	BESS Degradation	Uncertainty	Electric Vehicle
[17]	✓	×	×	✓	×
[18]	✓	×	×	×	✓
[19]	✓	×	×	×	×
[20]	✓	×	×	×	×
[21]	✓	×	×	×	×
[22]	✓	×	×	×	×
[23]	✓	×	×	×	×
[41]	✓	×	×	×	×
[24]	✓	×	✓	×	✓
[25]	✓	×	×	×	×
[28]	✓	×	×	×	×
[29]	✓	×	×	×	×
[26]	✓	✓	×	×	×
[27]	✓	✓	×	×	×
[31]	✓	✓	✓	×	×
[42]	✓	×	×	×	×
[43]	✓	✓	×	×	×
[34]	✓	×	✓	×	×
[35]	✓	×	✓	×	×
[36]	✓	×	✓	×	✓
[37]	✓	×	×	✓	×
[38]	✓	×	×	✓	×
[40]	✓	×	×	×	✓
Proposed EMS	✓	✓	✓	✓	✓

variability in arrival times, charging durations, and coordination with solar generation and grid tariffs.

- *Need for Holistic EMS Integration:* While individual studies address aspects like economic optimization, voltage regulation, or BESS degradation, very few EMS frameworks integrate all these components effectively. Existing literature reveals a gap in comprehensive EMS models that balance economic sustainability with robust voltage regulation and effective BESS degradation management under uncertain conditions.

The proposed EMS overcomes the limitations identified in the literature by incorporating multiple functionalities, delivering a comprehensive solution that balances economic optimization, voltage stability, and minimization of BESS degradation. Its key attributes are as follows:

- *Stochastic Model Predictive Control (SMPC) for BESS Operation:* The EMS employs a finite-horizon SMPC scheme to optimize BESS operation over a 24-hour prediction horizon. This approach aligns battery usage with solar generation, residential load demand, EV charging, electricity pricing, and feed-in tariffs. The SMPC framework not only manages uncertainties through scenario generation but also incorporates battery degradation into the optimization to minimize operational costs, extend battery lifespan, and ensure long-term sustainability. Additionally, the EMS supports voltage stability at the PCC by dynamically coordinating real and reactive power output, preventing voltage violations during solar peaks and peak load periods while reducing unnecessary BESS cycling.
- *Reactive Power Management for Voltage Stability:* The proposed EMS utilizes reactive power management via the BESS inverter to ensure PCC voltage remains within permissible limits by dynamically adjusting reactive power output in response to fluctuations in solar generation and load demand. This strategy alleviates stress on the BESS by minimizing reliance on real power adjustments, reducing unnecessary cycling, and extending battery life. It provides stability during both peak solar periods and high-demand intervals, maintaining reliable grid operations.
- *Degradation-Aware BESS Management for Long-Term Sustainability:* The proposed EMS integrates BESS degradation models within the optimization framework to enhance cost-effectiveness and operational longevity. This proactive management strategy reduces the frequency of replacements, extends the BESS lifespan, and ensures consistent performance, promoting sustainable energy management.
- *Uncertainty Management Through Scenario Generation:* The proposed EMS addresses variability in load demand, solar generation, and EV charging behavior through probabilistic scenario generation and detailed EV charging demand modeling. This enables the system to anticipate diverse conditions, ensuring reliable energy supply, optimized economic performance, and stable voltage regulation while effectively integrating renewable energy sources and managing grid constraints.
- *Comprehensive and Holistic Integration:* The proposed EMS offers a unified framework that integrates economic optimization, voltage stability, BESS degradation management, and uncertainty handling to deliver sustainable and robust energy management for residential complexes.

In summary, the proposed EMS enhances economic efficiency, ensures grid stability, and prolongs BESS lifespan, making it a significant advancement in sustainable residential energy management. A comparison of the attributes of existing EMSs with those of the proposed EMS is presented in Table 1.

2. Mathematical framework and configuration of a grid-integrated residential complex

In pursuit of carbon-neutral environments, residential complexes are increasingly adopting rooftop solar PV systems, BESS, and EVs. These complexes optimize economic benefits by managing local solar PV and BESS generation while considering grid electricity pricing. The emergence of EVs has shifted load demand profiles, introducing uncertainty due to the random nature of EV charging. To accommodate this, EV charging stations are integrated, promoting sustainable transportation and addressing residents' evolving energy needs. Combining rooftop solar installations and shared battery storage enhances energy efficiency, reduces grid dependence, and aligns with net-zero carbon emission goals. A schematic representation of a grid-connected residential complex, primarily powered by rooftop solar energy systems and supplemented with a BESS, is illustrated in Fig. 1.

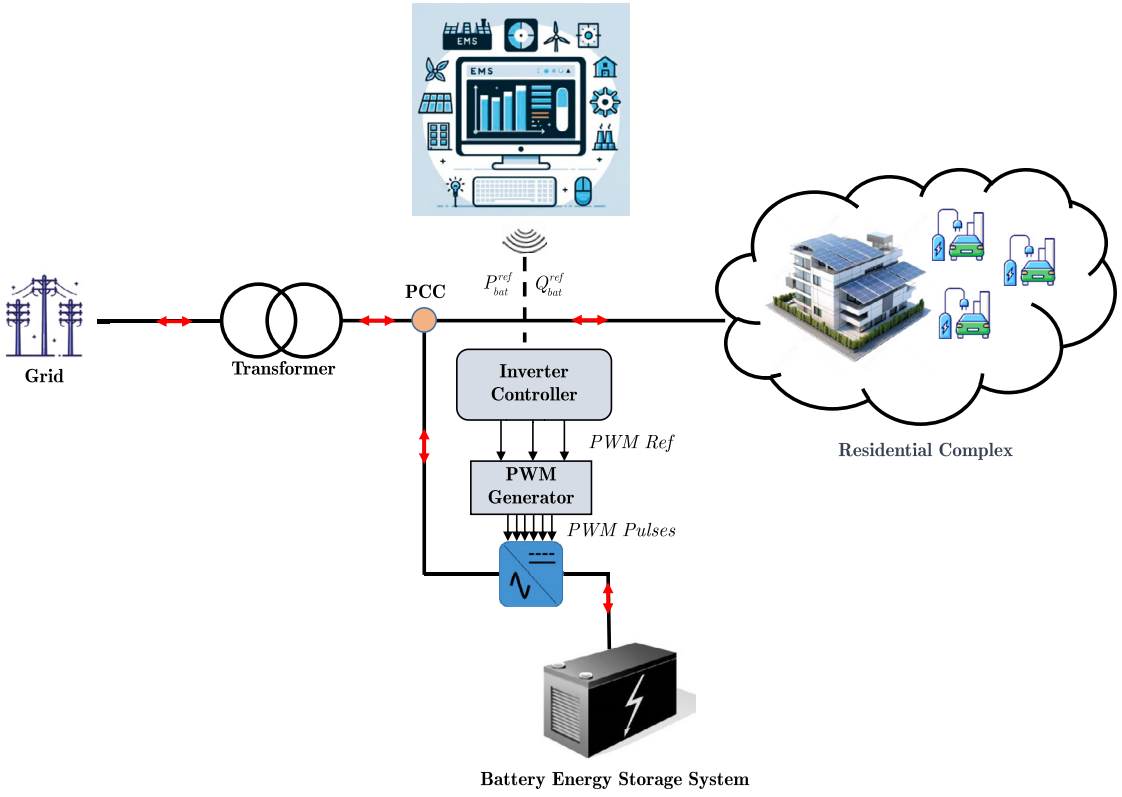


Fig. 1. Schematic Diagram of a Microgrid.

2.1. Rooftop solar PV systems

Rooftop solar PV systems are becoming essential components of residential households, serving as a primary source of renewable energy. The electrical power output from each rooftop solar installation at a time instance at the h_i^{th} hour $t_l \in h_i$, is dependent on the solar irradiation $G(t_l)$ and the capacity of the rooftop solar C_j. The power output can be modeled by accounting for the associated inverter efficiency η_j, and the temperature coefficient k, as well as the solar module temperature $T_{\mathrm{m}}(t_l)$ relative to the standard test condition temperature $T_{\mathrm{STC}} = 25\,^{\circ}\mathrm{C}$. This relationship is expressed as:

$$\mathrm{P}_{\mathrm{PV},j}(t_l) = \eta_j C_j \times G(t_l) \times \left(1 + k(T_{\mathrm{m}}(t_l) - T_{\mathrm{STC}})\right) \tag{1}$$

The total power output from all rooftop installations within the residential complex at time t_l is the sum of the outputs from n_{PV} individual rooftop solar installations:

$$\mathbf{P}_{\mathrm{PV}}(t_l) = \sum_{j=1}^{n_{PV}} \mathrm{P}_{\mathrm{PV},j}(t_l) \tag{2}$$

Assuming that the power generation remains constant within short intervals Δt within an hour, the harvested solar energy can be calculated as:

$$E_{\mathrm{PV}}(h_i) = \sum_{t_l \in h_i} \mathbf{P}_{\mathrm{PV}}(t_l)\Delta t \tag{3}$$

where Δt is the interval over which the power is assumed constant. To link the harvested energy with the average power output from the solar installation in the complex, we can use the following relationship:

$$E_{\mathrm{PV}}(h_i) \approx \overline{\mathbf{P}}_{\mathrm{PV}}(h_i) \times \Delta t (= 1\ \mathrm{hr}) \tag{4}$$

where $\overline{\mathbf{P}}_{\mathrm{PV}}(h_i)$ is the average power output within the h_i^{th} hour, and Δt is the duration of one hour. Due to the variability in weather conditions, there will be multiple possible scenarios of average solar power output at the h_i^{th} hour:

$$\left\{\overline{\mathbf{P}}_{\mathrm{PV}}^{(1)}(h_i), \overline{\mathbf{P}}_{\mathrm{PV}}^{(2)}(h_i), \ldots, \overline{\mathbf{P}}_{\mathrm{PV}}^{(S)}(h_i)\right\} \tag{5}$$

where $\overline{\mathbf{P}}_{\mathrm{PV}}^{(s)}(h_i)$ represent the average solar power during the h_i^{th} hour at the residential complex in the s-th scenario. Using these scenarios, one can estimate normal probability distribution of average solar power at that hour:

$$\mathbb{P}_{\overline{\mathbf{P}}_{\mathrm{PV}}}^{h_i} \sim \mathcal{N}\left(\mu_{\overline{\mathbf{P}}_{\mathrm{PV}}}^{h_i}, \left(\sigma_{\overline{\mathbf{P}}_{\mathrm{PV}}}^{h_i}\right)^2\right) \tag{6}$$

where $\mu_{\overline{\mathbf{P}}_{\mathrm{PV}}}^{h_i}$ and $\sigma_{\overline{\mathbf{P}}_{\mathrm{PV}}}^{h_i}$ are the mean and standard deviation of the average solar power output, respectively.

2.2. Residential community electricity demand

2.2.1. Residential electricity demand

Electricity demand in a residential complex is influenced by the diverse activities of its inhabitants, resulting in unique patterns of electricity usage. Typically, residential loads follow a predictable 24-hour cycle aligned with daily human activities, but there is some uncertainty due to factors like seasonal weather changes, daylight hours, occupancy variations, and appliance usage patterns. Using historical energy consumption data, the statistical inference of hourly average power demand can be determined:

$$\mathbb{P}_{\overline{\mathbf{P}}_{\mathrm{D,H}}}^{h_i} \sim \mathcal{N}\left(\mu_{\overline{\mathbf{P}}_{\mathrm{D,H}}}^{h_i}, \left(\sigma_{\overline{\mathbf{P}}_{\mathrm{D,H}}}^{h_i}\right)^2\right) \tag{7}$$

where $\mu_{\overline{\mathbf{P}}_{\mathrm{D,H}}}^{h_i}$ and $\sigma_{\overline{\mathbf{P}}_{\mathrm{D,H}}}^{h_i}$ are the mean and standard deviation of the average power demand during the h_i^{th} hour.

2.2.2. EV charging demand

EV charging demand adds complexity to the residential energy profile and cannot be easily modeled from historical data like regular load demand. This is due to the lack of readily available historical data and the variability in individual mobility habits, charging frequency, and timing. Unlike regular residential loads, EV charging demand is unpredictable and influenced by user behavior and peak hours when residents charge their vehicles. To address this, we introduce an EV charging demand model within the residential complex framework.

Consider a residential complex with n_H households, of which n_{EV} households own EVs. The complex is equipped with n_{CS} charging stations, where $n_{CS} < n_{EV} < n_H$. Each EV participates in a set of activities $\{a_p\}_{p=1}^{P}$ throughout the day. These activities include commuting to and from work, grocery trips, food pick-ups, school drop-offs and pick-ups, social visits, recreational activities, long-distance travel, and more. Each activity is associated with specific travel distances $\{d_p\}_{p=1}^{P}$. Probabilities $\{\pi_{h_i,p}\}_{p=1}^{P}$ are assigned to each activity for every hour of the day, including the probability of each EV being parked at the residential complex at a specific hour $\pi_{h_i,\text{park}}$. At any given hour, the sum of the probabilities of all activities, including parking, equals 1, ensuring that all possible states are accounted for:

$$\sum_{p=1}^{P} \pi_{h_i,p} + \pi_{h_i,\text{park}} = 1 \tag{8}$$

The loss in SoC for the e-th EV due to a particular activity a_p is given by:

$$\Delta\text{SoC}_e^{a_p} = -\frac{\lambda_e \times d_p}{C_e} \times 100 \tag{9}$$

where λ_e is the energy consumption rate of the EV in kWh per km driven, and C_e is the EV battery capacity in kWh. For each hour h_i, the expected SoC $\mathbb{E}[\text{SoC}_e(h_i)]$ for each EV $e \in n_{EV}$ is updated based on the probabilities of different activities and their corresponding SoC changes:

$$\mathbb{E}[\text{SoC}_e(h_{i+1})] = \mathbb{E}[\text{SoC}_e(h_i)] + \sum_{p=1}^{P} \pi_{h_i,p} \Delta\text{SoC}_e^{a_p} \tag{10}$$

The expected charging energy demand $\mathbb{E}[E_{EV,e}(h_i)]$ for each EV e at each hour h_i is given by:

$$\mathbb{E}[\text{E}_{\text{EV,e}}(h_i)] = \pi_{h_i,\text{park}} \times \min\left(\text{E}_{\text{draw},e}, \frac{100 - \mathbb{E}[\text{SoC}_e(h_i)]}{100} \times C_e\right) \times \mathbf{1}_{\{\mathbb{E}[\text{SoC}_e(h_i)]<100\}} \tag{11}$$

where $\text{E}_{\text{draw},e}$ is the energy that the e-th EV battery can draw from the charging station per hour, measured in kWh.

Given the limited number of charging stations (n_{CS}) compared to the number of EVs (n_{EV}) in the residential complex ($n_{CS} < n_{EV}$), not all EVs can charge simultaneously. To manage this constraint, a *first-come, first-serve (FCFS)* strategy is adopted, ensuring that EVs are prioritized based on their arrival times at the charging stations. This method reflects typical real-world behavior in shared infrastructure scenarios and is implemented as follows:

- Identify EVs needing charging at hour h_i: At each hour h_i, EVs whose SoC is below 100% are identified as needing charging: $\mathbb{E}[\text{SoC}_e(h_i)] < 100\%$.
- Sort EVs by arrival times: The identified EVs are prioritized based on their arrival order at the charging stations. EVs arriving earlier are given higher priority, ensuring fair access.
- Allow the first n_{CS} EVs to charge: The first n_{CS} EVs (according to arrival times) are allowed to charge, while others must wait until a charging station becomes available. The total charging demand at hour h_i is determined as:

$$\mathbb{E}[\text{E}_{\text{EV,total}}(h_i)] = \min\left(n_{CS} \times \text{E}_{\max}, \sum_{e=1}^{n_{EV}} \mathbb{E}[\text{E}_{\text{EV,e}}(h_i)]\right), \tag{12}$$

where $E_{\max}$ is the maximum energy each station can deliver per hour, measured in kWh.

If all stations are occupied, EVs that cannot charge during the current time slot will wait for the next available slot. In residential complexes, where vehicles are often parked overnight, there is usually enough flexibility to shift charging to later times based on station availability.

After determining the total energy demand, we need to update the SoC for each EV based on actual charging. The set of EVs that will be charged, denoted as $S_{EV2ch}^{h_i}$, is determined based on arrival time and station availability:

$$S_{EV2ch}^{h_i} = \{e \in n_{EV} \mid \text{EV}_e \text{ is selected for charging based on arrival time and station availability}\} \tag{13}$$

The SoC for each EV in $S_{EV2ch}^{h_i}$ is updated using the following equation:

$$\mathbb{E}[\text{SoC}_e(h_{i+1})] = \mathbb{E}[\text{SoC}_e(h_i)] + \min\left(\text{E}_{\text{draw},e}, \frac{100 - \mathbb{E}[\text{SoC}_e(h_i)]}{100} \times C_e\right) \times \mathbf{1}_{\{\mathbb{E}[\text{SoC}_e(h_i)]<100\}} \quad \forall e \in S_{EV2ch}^{h_i} \tag{14}$$

We consider that the charging rate of the EVs remains constant throughout the hour h_i, therefore the expected EV charging power over the span of the h_i^{th} hour can be written:

$$\mathbb{E}[\overline{\mathbf{P}}_{\text{D,EV}}(h_i)] = \mathbb{E}[\text{E}_{\text{EV,total}}(h_i)] \tag{15}$$

Given the uncertainties associated with the travel distance for each of the considered activities, there will be uncertainties around the expected EV charging demand. This uncertainty can be incorporated in the model by presenting the following statistical representation:

$$\mathbb{P}_{\overline{\mathbf{P}}_{\text{D,EV}}}^{h_i} \sim \mathcal{N}\left(\mu_{\overline{\mathbf{P}}_{\text{D,EV}}}^{h_i}, \left(\sigma_{\overline{\mathbf{P}}_{\text{D,EV}}}^{h_i}\right)^2\right) \tag{16}$$

where $\mu_{\overline{\mathbf{P}}_{\text{D,EV}}}^{h_i} = \mathbb{E}[\overline{\mathbf{P}}_{\text{D,EV}}(h_i)]$. The standard deviation $\sigma_{\overline{\mathbf{P}}_{\text{D,EV}}}^{h_i}$ accounts for the variability in travel distances and other stochastic factors affecting EV activities, such as different trip patterns on weekends and holidays, variations in daily travel distances, unexpected errands, and differences in charging behaviors among residents. It can be calculated by considering the variability in travel distances and their associated probabilities for each activity. Using the travel distances $\{d_p\}_{p=1}^{P}$ and their standard deviations $\{\sigma_p\}_{p=1}^{P}$, along with the activity probabilities $\{\pi_{h_i,p}\}_{p=1}^{P}$, we can aggregate the uncertainties to compute $\sigma_{\text{D,EV}}^{h_i}$. This accounts for the stochastic nature of EV activities and their impact on the charging demand.

2.3. Battery energy storage system (BESS) operation

BESS are becoming increasingly common in residential complexes, especially those with EV charging infrastructure. They enhance energy efficiency by managing the variability in rooftop solar PV generation, EV charging demands, and residential load demands. The BESS stores surplus energy during high solar production or low demand and discharges it during peak usage or low solar output. Additionally, the BESS can exchange power with the grid, selling excess energy at feed-in tariff rates and purchasing electricity when grid prices are favorable. This helps maintain a stable energy supply, optimizes the use of rooftop solar energy, and accommodates increasing EV charging needs.

A BESS operates in either discharge or charging mode, dynamically switching between these states based on energy supply and demand conditions. In discharging mode, the BESS supplies energy to the residential complex, reducing the stored energy as it meets the demand. Conversely,

in charging mode, the BESS stores excess energy from rooftop solar PV systems or the grid, increasing its stored energy. This dual functionality allows the BESS to stabilize the energy supply, manage peak loads, and optimize the use of harvested solar energy within the residential complex. The following model describes the operation of a BESS in a residential complex at any instance $t_l \in h_i$ within the hour h_i.

When the BESS is discharging, the available energy in the BESS at instance t_l is given by:

$$\mathrm{E_B}(t_l) = \mathrm{E_B}(t_{l-1}) - \Delta\mathrm{E_{B,Dis}}(t_l) \tag{17}$$

$$\Delta\mathrm{E_{B,Dis}}(t_l) = \frac{\mathrm{P_{B,Dis}}(t_l)}{\eta_\mathrm{B} \times 1000} \times \frac{\Delta t}{3600}; \quad \Delta t = t_l - t_{l-1} \tag{18}$$

On the other hand, the energy balance of the BESS during the charging mode at instance t_l is given as follows:

$$\mathrm{E_B}(t_l) = \mathrm{E_B}(t_{l-1}) + \Delta\mathrm{E_{B,Chg}}(t_l) \tag{19}$$

$$\Delta\mathrm{E_{B,Chg}}(t_l) = \mathrm{P_{B,Chg}}(t_l) \times \eta_\mathrm{B} \times \frac{\Delta t}{1000 \times 3600}; \quad \Delta t = t_l - t_{l-1} \tag{20}$$

where $\mathrm{E_B}(t_l)$ is the energy stored in the BESS at time t_l, $\Delta\mathrm{E_{B,Dis}}(t_l)$ represents the energy discharged, $\Delta\mathrm{E_{B,Chg}}(t_l)$ is the amount of energy input, $\mathrm{P_{B,Dis}}(t_l)$ is the power discharged by the BESS at time t_l, $\mathrm{P_{B,Chg}}(t_l)$ is the power used to charge the BESS at time t_l, and η_B is the efficiency of the BESS. The BESS SoC during both discharging and charging modes of operation can be computed as follows:

$$\text{Discharging: } \mathrm{SoC}(t_l) = \mathrm{SoC}(t_{l-1}) - \frac{\Delta\mathrm{E_{B,Dis}}(t_l)}{\mathrm{C_B}(t_l)} \times 100\% \tag{21}$$

$$\text{Charging: } \mathrm{SoC}(t_l) = \mathrm{SoC}(t_{l-1}) + \frac{\Delta\mathrm{E_{B,Chg}}(t_l)}{\mathrm{C_B}(t_l)} \times 100\% \tag{22}$$

The SoC is maintained within the limits:

$$\mathrm{SoC_{min}} \le \mathrm{SoC}(t_l) \le \mathrm{SoC_{max}} \tag{23}$$

The charging/discharging power of the BESS at any instance t_l is constrained as follows:

$$-\mathrm{P_B^{max}} \le \mathrm{P_B}(t_l) \le \mathrm{P_B^{max}} \tag{24}$$

where $\mathrm{P_B}(t_l)$ is the output (discharging)/ input(charging) power of the BESS at instance $t_l \in h_i$. The term $\mathrm{C_B}(t_l)$ represents the capacity of the BESS at that instance, which degrades over time. This degradation can be modeled using the following equation [44,45]:

$$\mathrm{E_{degrade}}(t_l) = \alpha\sqrt{\mathrm{C_N}}\, e^{\frac{-E_a}{R \cdot T(t_l)}} + \left(\gamma_0 + \frac{\gamma_1}{\mathrm{C_{Rate}}(t_l)} + \frac{\gamma_2}{\mathrm{C_{Rate}^2}(t_l)} + \frac{\gamma_3}{\mathrm{C_{Rate}^3}(t_l)}\right) \tag{25}$$

where α is degradation co-efficient associated with the cycle number, $\mathrm{C_N}$ is the nominal capacity of the BESS, E_a is the activation energy, R is the universal gas constant, $T(t_l)$ is the temperature at time t_l, $\gamma_0, \gamma_1, \gamma_2$, and γ_3 are coefficients for the C-rate dependency of BESS degradation, $\mathrm{C_{Rate}}(t_l)$ is the charge/discharge rate at time t_l. This degradation model combines a temperature and cycle count-based Arrhenius-type model with a polynomial dependency on the C-rate, reflecting the effects of operational stress on battery longevity. The reduced BESS storage capacity at instance t_l due to degradation is:

$$\mathrm{C_B}(t_l) = \mathrm{C_B}(t_{l-1}) - \mathrm{E_{degrade}}(t_l) \tag{26}$$

The inverter in the BESS is essential for converting DC power from the BESS to AC power for residential use and vice versa, enabling bidirectional power flow management within the residential complex. The power dynamics through the inverter can be described as follows:

$$\mathrm{P_{B,INV}}(t_l) = \eta_\mathrm{INV} \cdot \mathrm{P_B}(t_l) \tag{27}$$

where, $\mathrm{P_{B,INV}}(t_l)$ denotes the AC power output from the BESS through the inverter, $\mathrm{P_B}(t_l)$ represents the DC power output, and η_INV is the inverter efficiency. The inverter's capacity $\mathrm{S_{B,INV}}$ and its charging or discharging power $P_{\mathrm{B,INV}}(t_l)$ determine the reactive power capability $\mathrm{Q_{B,INV}}(t_l)$, constrained by:

$$|\mathrm{P_{B,INV}}(t_l)|^2 + |\mathrm{Q_{B,INV}}(t_l)|^2 \le \mathrm{S_{B,INV}^2} \tag{28}$$

2.4. Power flow and at the point of common coupling (PCC)

This connection point is vital for balancing power flows between the grid and various energy sources within the complex, such as rooftop solar PV systems, BESS, and EV charging stations. Effective management of the PCC ensures that real and reactive power exchanges are balanced, maintaining voltage levels within defined limits to provide a stable and reliable power supply. At an instance $t_l \in h_i$, the real ($\mathrm{P_{PCC}}(t_l)$) and reactive power ($\mathrm{Q_{PCC}}(t_l)$) flows at the PCC can be described as:

$$\mathrm{P_{PCC}}(t_l) = \mathrm{P_{PV}}(t_l) - \mathrm{P_{D,RC}}(t_l) + \mathrm{P_{B,INV}}(t_l) \tag{29}$$

$$\mathrm{Q_{PCC}}(t_l) = -\mathrm{Q_{D,RC}}(t_l) + \mathrm{Q_{B,INV}}(t_l) \tag{30}$$

where $\mathrm{P_{PV}}(t_l) = \overline{\mathbf{P}}_{\mathrm{PV}}^{h_i} + \delta_{\mathrm{PV}}^{h_i}$, $\overline{\mathbf{P}}_{\mathrm{PV}}^{h_i} \in \mathbb{P}_{\mathrm{PV}}^{h_i}$, and $\delta_{\mathrm{PV}}^{h_i}$ is the granular time scale uncertainty of harvested solar power around the hourly average harvested solar power in the residential complex. The term $\mathrm{P_{D,RC}}(t_l)$ represents one of the probable total real power demand scenarios at the residential complex at time $t_l \in h_i$, which is given by:

$$\mathrm{P_{D,RC}}(t_l) = \mathrm{P_{D,H}}(t_l) + \mathrm{P_{D,EV}}(t_l) \tag{31}$$

where $\mathrm{P_{D,H}}(t_l)$ and $\mathrm{P_{D,EV}}(t_l)$ are defined as: $\mathrm{P_{D,H}}(t_l) = \overline{\mathbf{P}}_{\mathrm{D,H}}^{h_i} + \delta_{\mathrm{D,H}}^{h_i}$ and $\mathrm{P_{D,EV}}(t_l) = \overline{\mathbf{P}}_{\mathrm{D,EV}}^{h_i} + \delta_{\mathrm{D,EV}}^{h_i}$. The terms $\overline{\mathbf{P}}_{\mathrm{D,H}}^{h_i}$ and $\overline{\mathbf{P}}_{\mathrm{D,EV}}^{h_i}$ are the average power demands for household activities and EV charging during the h_i^{th} hour, respectively, and $\delta_{\mathrm{D,H}}^{h_i}$ and $\delta_{\mathrm{D,EV}}^{h_i}$ are the uncertainties around these averages. The reactive power demand at the residential complex at instance t_l is determined according to the power factor of the aggregated load demand at the residential complex. The power exchange at the PCC ($\vec{\mathrm{S}}_{\mathrm{PCC}}(t_l) = \mathrm{P_{PCC}}(t_l) + j\mathrm{Q_{PCC}}(t_l)$) at the instance t_l determines the PCC voltage at that instance, which can be expressed as:

$$\vec{\mathrm{V}}_{\mathrm{PCC}}(t_l) = \vec{\mathrm{E}}_{\mathrm{Grid}}(t_l) + \vec{\mathrm{I}}_{\mathrm{PCC}}(t_l)\left(\mathrm{Z_{Grid}} + \mathrm{Z_{Feeder}}\right) = \vec{\mathrm{E}}_{\mathrm{Grid}}(t_l) + \vec{\mathrm{I}}_{\mathrm{PCC}}(t_l)\mathrm{Z_{GF}} \tag{32}$$

where $\vec{\mathrm{I}}_{\mathrm{PCC}}(t_l)$ is the current corresponding to PCC power $\vec{\mathrm{S}}_{\mathrm{PCC}}(t_l)$, $\mathrm{Z_{Feeder}} = \mathrm{R_{Feeder}} + j\mathrm{X_{Feeder}}$ is the impedance of the feeder and distribution transformer connecting the residential complex to the grid, the term $\mathrm{Z_{GF}} = \mathrm{Z_{Grid}} + \mathrm{Z_{Feeder}}$. The grid is modeled as an equivalent voltage source $\vec{\mathrm{E}}_{\mathrm{Grid}}$ behind an equivalent impedance $\mathrm{Z_{Grid}} = \mathrm{R_{Grid}} + j\mathrm{X_{Grid}}$. For the detailed derivation of PCC voltage refer to [46]. To ensure stable operation, the PCC voltage must be maintained within a defined range:

$$\mathrm{V}_{min} \le |\vec{\mathrm{V}}_{\mathrm{PCC}}(t_l)| \le \mathrm{V}_{max} \tag{33}$$

Managing energy in a residential complex with rooftop solar PV systems, BESS, and EV charging infrastructure requires an efficient EMS. The EMS optimizes the BESS's charging and discharging power and its reactive power output, adjusting for residential load demands, solar PV production, grid electricity pricing, and PCC voltage. By balancing energy flow at the PCC, the EMS ensures a stable energy supply, addressing variable residential and EV loads, fluctuating solar PV output, BESS health, and grid electricity price. In the following section, we propose an SMPC-based EMS that optimizes BESS operation in a residential complex. The EMS aims to maximize economic benefits by leveraging rooftop solar energy, addressing residential load demand, and managing EV charging. It also regulates PCC voltage by providing reactive power support from the BESS and minimizes BESS degradation.

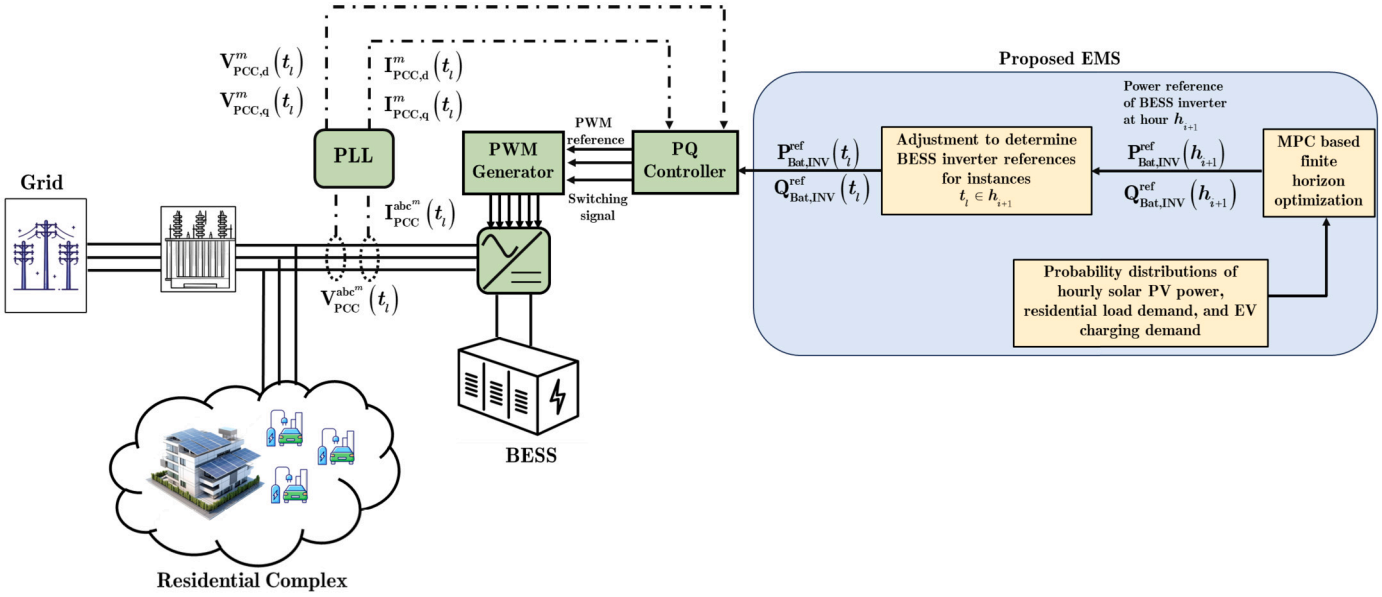


Fig. 2. A schematic diagram of the proposed EMS for BESS Regulation optimization in residential complexes.

3. Proposed EMS for optimizing BESS operation in residential complexes

3.1. Operational framework of the proposed EMS

The proposed EMS aims to (i) maximize economic benefits by utilizing solar PV energy, (ii) minimize battery degradation, and (iii) ensure voltage stability at the PCC while accounting for uncertainties in solar PV output and load demand. This study does not aim to fully satisfy all EV charging demand. Instead, EV charging demand is modeled as part of the overall load profile, contributing to the broader energy management problem. The EMS optimizes the use of solar PV and battery resources to minimize grid dependence and ensure operational stability, rather than guaranteeing that all EVs are charged at all times. Such behavior-oriented optimizations, such as dynamically managing queuing or availability, are beyond the scope of this study. A schematic of the proposed EMS and its interaction with the energy systems in a residential complex has been shown in Fig. 2. The EMS operates hourly, at the h_i^{th} hour, by generating power references for the following hour to manage BESS operations, aligning with predicted energy demands and solar generation. It uses a finite horizon SMPC approach to optimize BESS power output over 24 hours, considering expected PV generation, residential and EV demand, and fluctuating electricity prices. At each operating hour, the EMS optimizes the operation of the BESS in the residential complex by accounting for a set of probable operating scenarios that reflect the uncertainties in load demand, solar PV generation, and EV charging demand. The hourly work-flow of the proposed SMPC based EMS has been presented next.

Step 1: Deriving 24-Hour Ahead Operating Scenarios: First, during the operating hour h_i in the proposed EMS, a set of probable 24-hour ahead operating scenarios is generated. These scenarios are based on the probability distributions of average hourly harvested solar power, residential load demand, and EV charging demand for the next 24 hours, as defined earlier in the paper ((6), (7), and (16)). These operating scenarios are created by drawing correlated random samples from the corresponding probability distributions, resulting in the set of scenarios:

$$\mathbf{OS}_{h_i}^{24} = \{\mathbf{OS}_1, \ldots, \mathbf{OS}_N\} \tag{34}$$

where $\mathbf{OS}_{h_i}^{24}$ represents the set of N probable operating scenarios for the next 24 hours. Each scenario $\mathbf{OS}_n$ $(n = 1, \ldots, N)$ consists of correlated random samples of average hourly load demand, solar power, and EV charging demand for each hour h_{i+p} $(p = 1, \ldots, 24)$ within the 24-hour period. The average load demand, solar power, and EV charging data for each hour are correlated with those of the other hours. Additionally, the average hourly solar generation and load demand are correlated to reflect the interdependencies across the 24-hour period.

$$\mathbf{OS}_n = \left\{ \left(\overline{\mathbf{P}}_{\mathrm{D,H},n}^{h_{i+p}}, \overline{\mathbf{P}}_{\mathrm{PV},n}^{h_{i+p}}, \overline{\mathbf{P}}_{\mathrm{D,EV},n}^{h_{i+p}} \right) \mid p = 1, \ldots, 24 \right\} \tag{35}$$

In the Appendix A section, we provide a detailed explanation of how these operating scenarios are generated for a particular hour h_i.

Step 2: Formulate Optimization Problem: At the ith operating hour, the EMS aims to minimize the expected total operational costs comprising energy cost and battery degradation cost over the next 24-hour prediction horizon, considering the set of N probable scenarios in $\mathbf{OS}_{h_i}^{24}$. The optimization problem is formulated as follows:

$$\underset{\left\{\overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{h_{i+p}}, \overline{\mathbf{Q}}_{\mathrm{Bat,INV}}^{h_{i+p}}\right\}_{p=1}^{24}}{\text{Minimize}} \quad \mathbb{E}(\mathrm{C}_{24hr}) = \frac{1}{N} \sum_{n=1}^{N} \left(\gamma \times \mathrm{COE}_n + \delta \times \mathrm{COD}_n \right) \tag{36}$$

Subject to:

$$\text{BESS Constraints: } \mathrm{SOC}_{\min} \leq \mathrm{SOC}_n(t_l) \leq \mathrm{SOC}_{\max}, \quad \forall n, p, t_l \in h_{i+p} \tag{37}$$

Algorithm 1 Work-flow of the proposed EMS at the h_i^{th} of the day.

Require: Probability distributions of hourly average solar PV power $\mathbb{P}^{h_i}_{\overline{\mathbf{P}}_{\mathrm{PV}}}$, residential load demand $\mathbb{P}^{h_i}_{\overline{\mathbf{P}}_{\mathrm{D,H}}}$, and EV charging demand $\mathbb{P}^{h_i}_{\overline{\mathbf{P}}_{\mathrm{D,EV}}}$, which have been defined in (6), (7), and (16), respectively.

1: Generate N probable 24-hour ahead operating scenarios by drawing correlated random samples from the probability distributions of average hourly solar power, residential load demand, and EV charging demand.
2: $\mathbf{OS}_{h_i}^{24} \leftarrow \{\mathbf{OS}_1, \mathbf{OS}_2, \ldots, \mathbf{OS}_N\}$
3: **for** $h \leftarrow 1$ to 24 **do**
4: Solve the SMPC optimization problem at h-th hour using eqs. (36)-(41)
5: $\left\{\overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{h^*_{i+p}}, \overline{\mathbf{Q}}_{\mathrm{Bat,INV}}^{h^*_{i+p}}\right\}, \cdots, \left\{\overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{h^*_{i+24}}, \overline{\mathbf{Q}}_{\mathrm{Bat,INV}}^{h^*_{i+24}}\right\} \leftarrow$ Optimal sequences of BESS output power for next 24 hours
6: **end for**
7: Use the optimal BESS output power $\overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{hi+1}$ and $\overline{\mathbf{Q}}_{\mathrm{Bat,INV}}^{hi+1}$ determined by SMPC for the $(i+1)$-th hour as the power reference for the BESS inverter controller.

$$-\mathrm{P}_{\mathrm{Bat}}^{\max} \leq \overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{h_{i+p}} \leq \mathrm{P}_{\mathrm{Bat}}^{\max}, \quad \forall n, p \tag{38}$$

$$\text{Inverter Constraint: } |\overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{h_{i+p}}|^2 + |\overline{\mathbf{Q}}_{\mathrm{Bat,INV}}^{h_{i+p}}|^2 \leq \mathrm{S}_{\mathrm{B,INV}}^2, \quad \forall n, p \tag{39}$$

$$\text{PCC Voltage Constraint: } \mathrm{V}_{\min} \leq |\vec{\mathrm{V}}_{\mathrm{PCC},n}(t_l)| \leq \mathrm{V}_{\max}, \quad \forall n, p, t_l \in h_{i+p} \tag{40}$$

In this problem, $\mathbb{E}(\mathrm{C}_{24hr})$ represents the expected total operational cost over 24 hours. The cost of energy COE_n for the nth scenario is calculated as:

$$\mathrm{COE}_n = \sum_{p=1}^{24} \left(\mathrm{E}_{\mathrm{Buy},n}(h_{i+p}) \times \mathrm{Buy}_{\mathrm{Price}}(h_{i+p}) - \mathrm{E}_{\mathrm{Sell},n}(h_{i+p}) \times \mathrm{Sell}_{\mathrm{Price}}(h_{i+p})\right) \tag{41}$$

The terms $\mathrm{Buy}_{\mathrm{Price}}(h_{i+p})$ and $\mathrm{Sell}_{\mathrm{Price}}(h_{i+p})$ are the electricity price and feed-in tariff at hour h_{i+p}, respectively. The energy sold and bought at hour h_{i+p} for scenario $\mathbf{OS}_n$, denoted as $\mathrm{E}_{\mathrm{Sell},n}(h_{i+p})$ and $\mathrm{E}_{\mathrm{Buy},n}(h_{i+p})$, are determined by:

$$\overline{\mathbf{P}}_{\mathrm{net},n}^{h_{i+p}} = \overline{\mathbf{P}}_{\mathrm{PV},n}^{h_{i+p}} + \overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{h_{i+p}} - \overline{\mathbf{P}}_{\mathrm{D,H},n}^{h_{i+p}} - \overline{\mathbf{P}}_{\mathrm{D,EV},n}^{h_{i+p}} \tag{42}$$

If $\overline{\mathbf{P}}_{\mathrm{net},n}^{h_{i+p}} > 0$:

$$\mathrm{E}_{\mathrm{Sell},n}(h_{i+p}) = \max\left(\frac{\overline{\mathbf{P}}_{\mathrm{net},n}^{h_{i+p}} \times \Delta h}{1000}, 0\right) \text{ kWh}; \quad \Delta h = h_{i+1} - h_i = 1 \text{ hr} \tag{43}$$

and

$$\mathrm{E}_{\mathrm{Buy},n}(h_{i+p}) = 0 \tag{44}$$

If $\overline{\mathbf{P}}_{\mathrm{net},n}^{h_{i+p}} < 0$:

$$\mathrm{E}_{\mathrm{Buy},n}(h_{i+p}) = \max\left(-\frac{\overline{\mathbf{P}}_{\mathrm{net},n}^{h_{i+p}} \times \Delta h}{1000}, 0\right) \text{ kWh} \tag{45}$$

and

$$\mathrm{E}_{\mathrm{Sell},n}(h_{i+p}) = 0 \tag{46}$$

The cost of battery degradation COD_n for the nth scenario is given by:

$$\mathrm{COD}_n = \sum_{p=1}^{24} \mathrm{E}_{\mathrm{deg},n}(h_{i+p}) \times \mathrm{Bat}_{\mathrm{Price/kWh}} \tag{47}$$

where $\mathrm{E}_{deg,n}(h_{i+p})$ denotes the degradation of the BESS at hour h_{i+p} in the nth scenario, which can be calculated using (25). The weighting factors γ and δ (with $\gamma + \delta = 1$) in the formulated optimization problem (36) balance energy costs and battery degradation. Adjusting these factors allows the operator to fine-tune the trade-off between minimizing energy expenses and reducing battery degradation.

Step 3: Solving the Optimization Problem: The formulated optimization problem, which includes minimizing expected total operational costs subject to BESS, inverter, and voltage constraints, can be approached using non-linear programming (NLP) techniques. Generally, such optimization problems are solved by defining the objective function and constraints, choosing an appropriate solver, initializing with a guess for the decision variables, and iterating until convergence criteria are met. In this work, we utilized MATLAB's *GlobalSearch* algorithm to solve the optimization problem. The selection of the GlobalSearch algorithm is warranted by the non-linear, multi-variable nature of the optimization problem embedded within the proposed EMS. The EMS incorporates multiple scenarios to account for uncertainties in solar generation, residential load demand, and EV charging, resulting in a non-convex solution space with non-linear constraints that require efficient management. These constraints such as battery SoC limits, inverter capacity, and PCC voltage stability further increase the complexity, making it challenging for local optimization methods to reliably identify the global optimum, as they may converge to suboptimal solutions. GlobalSearch addresses this challenge by exploring multiple starting points across the solution space and employing the interior-point algorithm within *fmincon* to find local optima, combining these results to approximate the global optimum. This approach enhances the robustness of the optimization process, ensuring that the EMS consistently identifies optimal control strategies across various operating scenarios. Although the GlobalSearch algorithm introduces computational overhead, the EMS mitigates this through an hourly optimization strategy. Executing the optimization once per hour balances computational feasibility with operational adaptability, allowing sufficient time to solve the complex optimization problem. This framework ensures the EMS can update its control actions based on new forecasts and evolving conditions, thereby maintaining efficient and reliable system operation.

Step 4: Implementing the Optimized Power Reference of the BESS Controller: The solution of the optimization problem yields the optimal control sequence for the BESS, providing both real and reactive power profiles for the upcoming 24-hour period, as detailed in the following expression:

$$\left\{\overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{h^*_{i+p}}, \overline{\mathbf{Q}}_{\mathrm{Bat,INV}}^{h^*_{i+p}}\right\}, \cdots, \left\{\overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{h^*_{i+24}}, \overline{\mathbf{Q}}_{\mathrm{Bat,INV}}^{h^*_{i+24}}\right\} \tag{48}$$

Once the 24-hour control sequence for the BESS has been established, the EMS proceeds to configure the BESS inverter controller power settings for the forthcoming hour ($i + 1^{th}$ hour). This is accomplished by setting the power references for the BESS inverter controller according to the optimized values obtained from the global optimization process. Specifically, the references of are set as follows:

$$\mathrm{P}_{\mathrm{Bat,INV}}^{\mathrm{ref}}(h_{i+1}) = \overline{\mathbf{P}}_{\mathrm{Bat,INV}}^{h^*_{i+1}}, \tag{49}$$

$$\mathrm{Q}_{\mathrm{Bat,INV}}^{\mathrm{ref}}(h_{i+1}) = \overline{\mathbf{Q}}_{\mathrm{Bat,INV}}^{h^*_{i+1}}. \tag{50}$$

Setting the power references for only the forthcoming hour ensures that the system can adapt to any changes or unforeseen events that may occur, maintaining flexibility and robustness in real-time operation. This hourly adjustment allows the EMS to incorporate the latest information and forecasts, enhancing overall system performance and reliability. An algorithm showing the work-flow of the EMS at the h_i^{th} hour has been presented in Algorithm 1.

The proposed EMS optimizes both real and reactive power to manage BESS operations, diverging from conventional approaches that focus only on real power for charging and discharging. By integrating reactive power management, the EMS directly addresses voltage stability at the PCC, which becomes critical during peak solar export and high-demand periods when fluctuations in power exchanges with the grid can affect voltage levels, especially in weak grid scenario. The optimization problem models the PCC voltage as a function of the power exchanged between the residential complex and the utility grid, incorporating specific constraints to ensure stable operation. These constraints, along with inverter capacity limits, ensure the optimization of real and reactive power outputs of the BESS while maintaining safe operational boundaries of the system. This formulation introduces additional complexity, as it requires simultaneous balancing of power flows, SoC limits, and voltage constraints. To enhance robustness, the EMS employs a scenario-based optimization approach to account for uncertainties in solar generation, load demand, and EV charging. Multiple scenarios reflect potential operating conditions, allowing the EMS to remain adaptive and effective in real-time operation. The scenario-based design ensures consistent PCC voltage regulation while optimizing economic performance and minimizing battery degradation. By integrating real and reactive power optimization with scenario-based planning, the proposed EMS provides a comprehensive energy management solution. This approach not only enhances voltage regulation but also maximizes cost savings and extends BESS lifespan through efficient operation under diverse conditions.

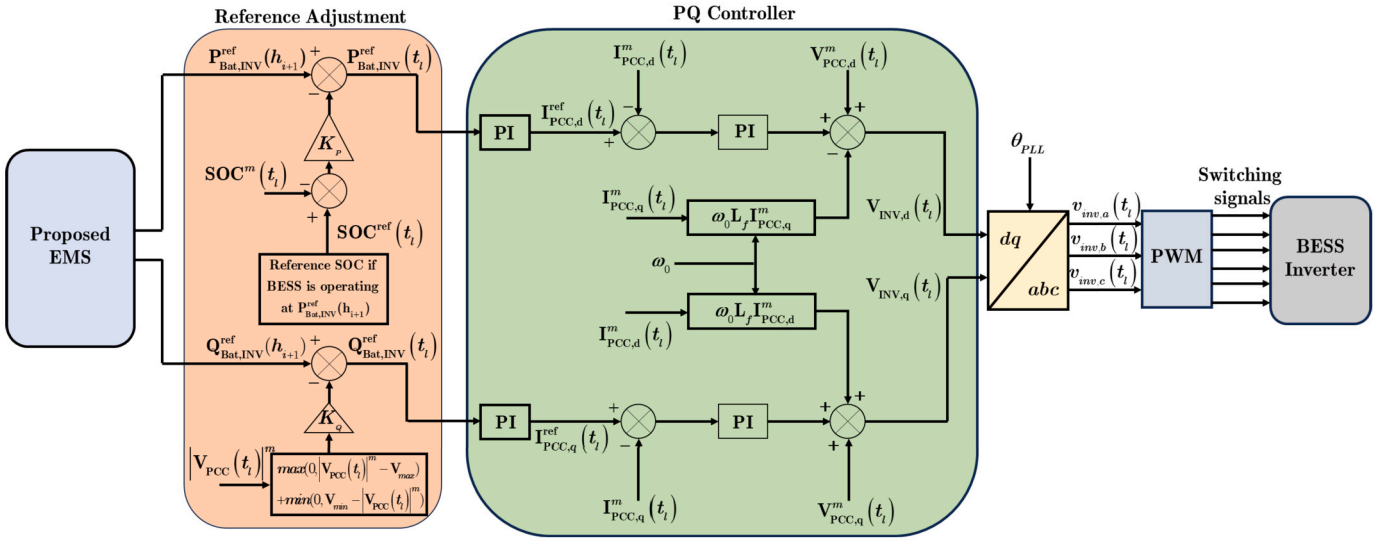


Fig. 3. Schematic diagram showing the interface between the proposed EMS and the BESS inverter P-Q controller.

3.2. Interface between the proposed EMS and the BESS inverter

As highlighted while describing the workflow of the proposed EMS, it operates on an hourly basis, whereas the BESS inverter functions on an instantaneous basis. In this section, we present how the proposed EMS is interfaced with the BESS inverter. It should be noted that the proposed EMS, while optimizing, considers hourly average solar power generation and household and EV charging demand and generates hourly references for BESS power. However, to accommodate load demand variability and fluctuations in solar PV generation within an hour, the hourly power references ($P^{ref}_{Bat,INV}(h_{i+1})$ and $Q^{ref}_{Bat,INV}(h_{i+1})$) derived by the proposed EMS are adjusted as below for any instance $t_l \in h_{i+1}$:

$$P^{ref}_{Bat,INV}(t_l) = P^{ref}_{Bat,INV}(h_{i+1}) - K_P\left(SOC(t_l)^{ref} - SOC(t_l)^m\right) \quad (51)$$

$$Q^{ref}_{Bat,INV}(t_l) = Q^{ref}_{Bat,INV}(h_{i+1}) - K_Q\Big(\max(0, |\vec{V}_{PCC}(t_l)^m| - V_{max}) + \min(0, V_{min} - |\vec{V}_{PCC}(t_l)^m|)\Big) \quad (52)$$

Here, K_P and K_Q are gain constants used to adjust the references. The term $SOC(t_l)^{ref}$ in (51) represents the target SOC for the BESS at time t_l, set to align with the real power output reference $P^{ref}_{Bat,INV}(h_{i+1})$ determined by the EMS, $SOC(t_l)^m$ is the measured BESS SOC at instant t_l. In (52), the term $|\vec{V}_{PCC}(t_l)^m|$ represents measured PCC voltage magnitude at time t_l.

Adjusting the reference power $P^{ref}_{Bat,INV}(t_l)$ based on the SOC is crucial to avoid overcharging or deep discharging of the battery, optimizing battery usage, and dynamically responding to real-time changes in solar generation and load demand. By keeping the SOC within a desired range, the BESS can effectively respond to variations in energy flow, ensuring reliable operation and extended battery life. Similarly, adjusting the reference reactive power $Q^{ref}_{Bat,INV}(t_l)$ based on the PCC voltage is essential for voltage regulation. Variations in load demand and solar generation can cause voltage fluctuations at the PCC. By adjusting $Q^{ref}_{Bat,INV}(t_l)$, the BESS can provide reactive power support to maintain voltage stability, ensuring the voltage remains within the desired range ($[V_{min} V_{max}]$).

The updated power references $P^{ref}_{Bat,INV}(t_l)$ and $Q^{ref}_{Bat,INV}(t_l)$ are utilized as inputs for the P-Q controller of the BESS inverter. The P-Q controller translates these power references into switching sequences to control the BESS inverter, ensuring that the BESS operates according to the adjusted power references. Various P-Q control strategies for BESS inverters have been extensively studied, including PI Control [47,48], PR Control [49], Vector Control [50,51], Space Vector Modulation (SVM) [52], and Sliding Mode Control (SMC) [53]. A schematic diagram illustrating the interface between the proposed EMS and the BESS inverter P-Q controller (employing the vector control approach [50]) is shown in Fig. 3.

4. Results and discussion

4.1. Test residential complex and simulation set-up

This section introduces case studies on a grid-connected residential complex to assess the effectiveness of the proposed EMS. The complex includes 100 homes, each with rooftop solar installations totaling 500 kW in capacity, supported by a 300 kWh lithium-ion BESS. The connected utility grid is considered to be relatively weak, with a short-circuit ratio (SCR) of 2.5, and the simulations are conducted using MATLAB Simulink via the Simscape Electrical toolbox. The proposed EMS is implemented as a MATLAB script, and its operations are simulated within MATLAB Simulink. A synchronization mechanism between the script and the Simulink model ensures seamless data flow and execution timing.

The load profiles are based on average hourly electricity usage data for three-bedroom town-houses in the Brisbane CBD, while solar output is calculated from local solar irradiation data. Based on these data for three years probability distributions of the hourly average residential power demand and harvest solar power are determined. In this simulation study, we consider that 30% of the 100 town-house residents own EVs, resulting in 30 EVs in the residential complex. The complex is equipped with 10 charging stations, each has a charging rate of 7.2 kW. The EVs have battery capacities ranging from 40 to 60 kWh and efficiencies between 6 and 8 km per kWh. Daily travel activities such as commuting, grocery trips, food pick-ups, and other errands are modeled based on typical residential patterns, with random assignment of times and distances. Table 2 presents the range of travel distances for the considered activities by EV owners. Different activity probabilities are assigned for weekdays, weekends, and holidays, influencing travel behavior, as shown in Tables 3, 4, and 5, respectively. Based on the above scenario, the charging demand at the EV charging stations is determined using the EV charging demand model presented in Section 2.2.2. Fig. 4 illustrates (a) the hourly EV charging demand at the residential complex over one year, and (b) the hourly EV charging demand over one specific week during the year. When evaluating the proposed EMS in this simulation study, various pricing models, such as fixed and Time-of-Use (ToU) rates from AGL Energy, including the solar savers feed-in tariff, were considered. Fig. 5 shows the hourly TOU electricity rates

Table 2
Travel distance range for different activities.

Activity	Going to workplace	Coming back from workplace	Grocery Trip	Other Errands	Long Distance Travel
Distance Range (km)	16-32	16-32	3-15	5-15	80-100

Table 3
Weekday probabilities of the considered activities.

Hour Interval	Going to Office	Coming Back from Office	Grocery Trip	Other Errands	Long Travel	Parked at Workplace	Parked at Home
00:00 - 04:00	0.01	0.00	0.01	0.01	0.01	0.00	0.96
04:00 - 08:00	0.05	0.00	0.02	0.02	0.02	0.00	0.89
08:00 - 12:00	0.20	0.00	0.05	0.05	0.05	0.50	0.15
12:00 - 16:00	0.10	0.00	0.10	0.10	0.10	0.40	0.20
16:00 - 20:00	0.00	0.30	0.10	0.10	0.10	0.00	0.40
20:00 - 24:00	0.00	0.10	0.05	0.05	0.05	0.00	0.75

Table 4
Weekend probabilities of the considered activities.

Hour Interval	Going to Office	Coming Back from Office	Grocery Trip	Other Errands	Long Travel	Parked at Workplace	Parked at Home
00:00 - 04:00	0.05	0.00	0.05	0.05	0.10	0.00	0.75
04:00 - 08:00	0.05	0.00	0.05	0.05	0.10	0.00	0.75
08:00 - 12:00	0.10	0.00	0.10	0.10	0.10	0.00	0.60
12:00 - 16:00	0.10	0.00	0.10	0.10	0.10	0.00	0.60
16:00 - 20:00	0.10	0.00	0.20	0.20	0.20	0.00	0.20
20:00 - 24:00	0.10	0.00	0.05	0.05	0.10	0.00	0.70

Table 5
Holiday probabilities of the considered activities.

Hour Interval	Going to Office	Coming Back from Office	Grocery Trip	Other Errands	Long Travel	Parked at Workplace	Parked at Home
00:00 - 04:00	0.05	0.00	0.05	0.05	0.10	0.00	0.75
04:00 - 08:00	0.05	0.00	0.05	0.05	0.10	0.00	0.75
08:00 - 12:00	0.10	0.00	0.10	0.10	0.10	0.00	0.60
12:00 - 16:00	0.10	0.00	0.10	0.10	0.10	0.00	0.60
16:00 - 20:00	0.10	0.00	0.20	0.20	0.20	0.00	0.20
20:00 - 24:00	0.10	0.00	0.05	0.05	0.10	0.00	0.70

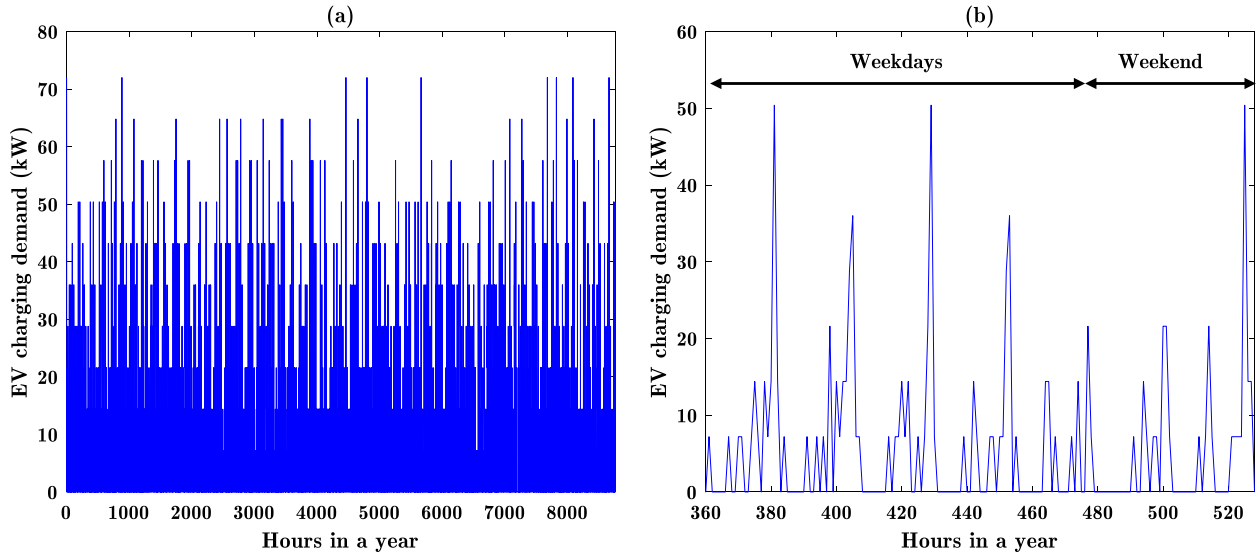


Fig. 4. (a) Hourly EV charging demand at the residential complex over one year, (b) Hourly EV charging demand over one week during the year.

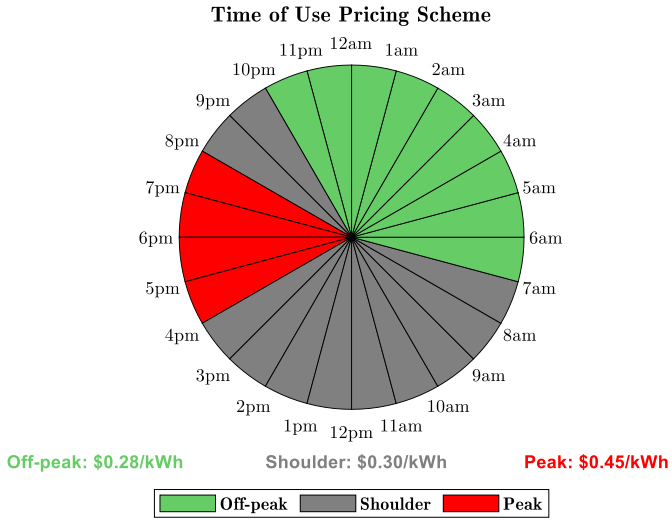


Fig. 5. Pricing structure considered in the case study. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

considered in this study, while fixed rate electricity price is $0.2/kWh, and feed-in tariff is $0.05/kWh.

4.2. Performance evaluation of the proposed EMS under fixed electricity pricing scheme

The proposed EMS has been evaluated over a one-year period under a fixed electricity pricing scheme. To demonstrate its performance, we present the energy distribution within the test residential complex over five consecutive summer days, as shown in Fig. 6. Fig. 6 (a) illustrates the dynamic interactions among fluctuating residential load, solar power generation, EV charging, and net power transactions with the utility grid, showcasing the EMS's ability to manage diverse operational conditions effectively. Fig. 6 (b) highlights the BESS operation, displaying its charging and discharging activities along with the reactive power contributions from the BESS inverter. This operation aims to maximize the economic benefits for the residential complex while ensuring PCC voltage stability and minimizing BESS degradation. These figures demonstrate the EMS's capability to regulate BESS operation, aligning it with residential and EV charging load demand, and harvested solar power. This optimization maximizes economic benefits by efficiently utilizing harvested solar power and minimizing BESS degradation during charging and discharging processes. During the solar hours, the EMS ensures that the BESS is fully charged with the surplus harvested solar power remaining after meeting the residential and EV charging load demands within the complex. The charging rate of the BESS is controlled to reach full capacity by the end of the solar hours, minimizing degradation. This is crucial since the degradation rate of the BESS is proportional to the charging and discharging rates, with controlled rates reducing wear and extending battery life. During the evening and night hours, when the residential and EV charging load demands increase, the EMS utilizes the stored solar power in the BESS to offset energy purchased from the grid to meet the load demand. The BESS discharges faster during the peak demand period and relatively slower after the peak demand period ends, maintaining the SoC within the limit ($20\% - 80\%$). Additionally, the EMS ensures the PCC voltage is maintained within the acceptable range of 0.95-1.05 pu by providing reactive power support from the BESS inverter. Specifically, during peak solar hours when exported solar power to the grid is maximum, the BESS inverter absorbs reactive power to ensure the voltage swelling at the PCC remains within the prescribed limit. Conversely, during the evening peak demand period, when the imported grid power is high, the BESS inverter supplies reactive power to prevent the PCC voltage from falling below the prescribed boundary. Fig. 7 (a) and (b) demonstrate PCC voltage over the span of the considered five summer days and for the months of January - February (summer months). For a more detailed evaluation, we present the operation of day 3 of selected summer days as shown in Fig. 8. These illustrations confirm the capability of the proposed EMS to adaptively modify the BESS's real and reactive power outputs based on solar availability, maximizing the economic benefit of the residential complex while ensuring PCC voltage stays within safe limits and minimizing BESS degradation.

Moving to the winter operational scenario, it is evident that the proposed EMS effectively adapts to seasonal variations. Fig. 9 illustrates the EMS's performance in managing the energy balance within the test residential complex over five representative winter days. This figure highlights how the EMS responds to variations in load demand, EV charging, diminished solar production, and interactions with the utility grid. During these winter days, the EMS dynamically adjusts the charging and discharging process of the BESS, similar to the process described for summer operation, to maximize economic benefits while minimizing BESS degradation.

One key aspect observed on day 5 of the winter period is the significantly lower harvested solar power. On this day, the EMS ensures that during solar hours, the surplus solar power, after meeting residential and EV charging load demand, is used to charge the BESS at a controlled rate instead of exporting to the grid. Consequently, the surplus solar energy is completely stored in the BESS, and its SoC reaches 64%, as the available solar energy is insufficient to fully charge the BESS. Since no power is exported to the grid during solar hours, there is no need for reactive power support from the BESS inverter to maintain PCC voltage. A similar behavior is seen during the solar hours of day 3 when very little power is exported, eliminating the need for reactive power absorption to maintain the PCC voltage. However, on days 1, 2, and 4, the BESS inverter provides reactive power absorption support to maintain the PCC voltage below the maximum allowable voltage with increased export of surplus solar energy after meeting residential and EV charging demand and fully charging the BESS. During the evening peak load demand period of each considered day, the EMS manages the drop in PCC voltage due to the increased import of grid power to meet residential and EV charging demand. The voltage profiles at the PCC throughout these winter days, depicted in Fig. 10, confirm that the EMS consistently maintains PCC voltage within the desired thresholds. These results demonstrate the capacity of the EMS to handle winter-specific conditions effectively, ensuring stable voltage levels and efficient energy management.

4.3. Performance evaluation of the proposed EMS under ToU electricity pricing scheme

In this section, we assess the effectiveness of the proposed EMS under the ToU electricity pricing scheme. First, we present the operation of the EMS in Fig. 11 for the same set of five consecutive summer days considered during the fixed-rate pricing scheme. This provides insights into how the proposed EMS adapts to varying energy prices. The figure highlights the EMS's ability to optimize BESS charging and discharging, as well as provide reactive power support, aiming to maximize economic benefits, minimize BESS degradation, and maintain PCC voltage within prescribed limits. The BESS undergoes charging during the shoulder electricity period (7 am - 4 pm), with the charging rate adjusted based on available solar energy for each respective day. The discharge cycle of the BESS aligns with peak electricity price periods (4 pm - 8 pm). This strategic coordination with ToU pricing allows the methodology to effectively utilize ToU pricing, thereby maximizing the economic benefits of the residential complex. Additionally, the EMS ensures that the BESS charging process is optimized to fully utilize available solar energy during the shoulder period, achieving full charge while minimizing BESS degradation. The EV charging demand is managed similarly to the fixed-rate scheme, integrating seamlessly with the overall energy management strategy. Reactive power support from the BESS inverter

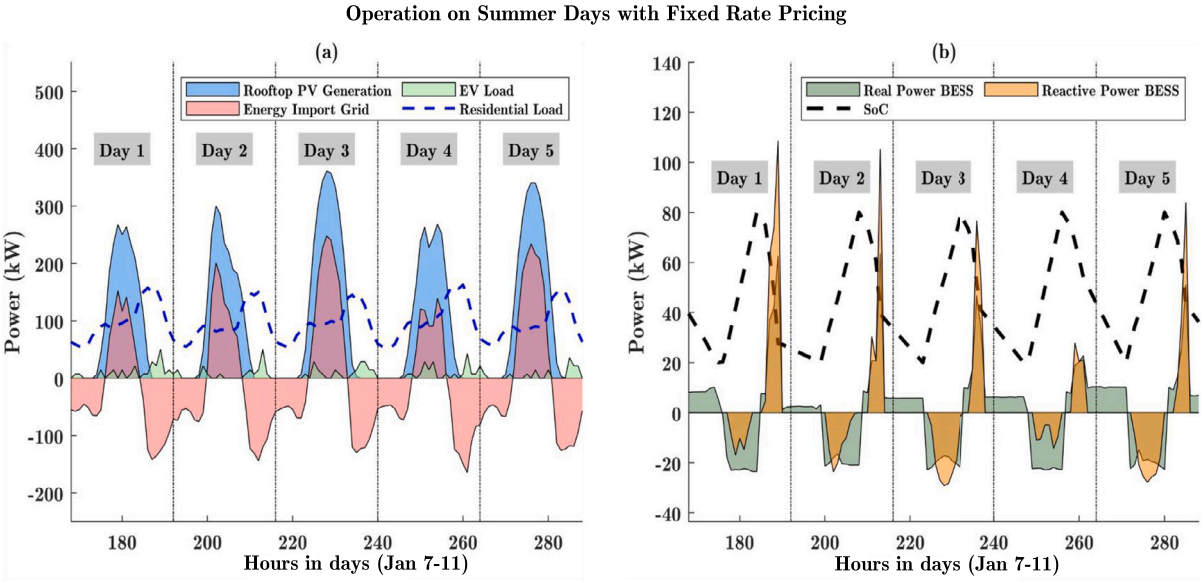


Fig. 6. Energy balance of the test residential complex for five consecutive summer days with fixed rate electricity pricing scheme. (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS.

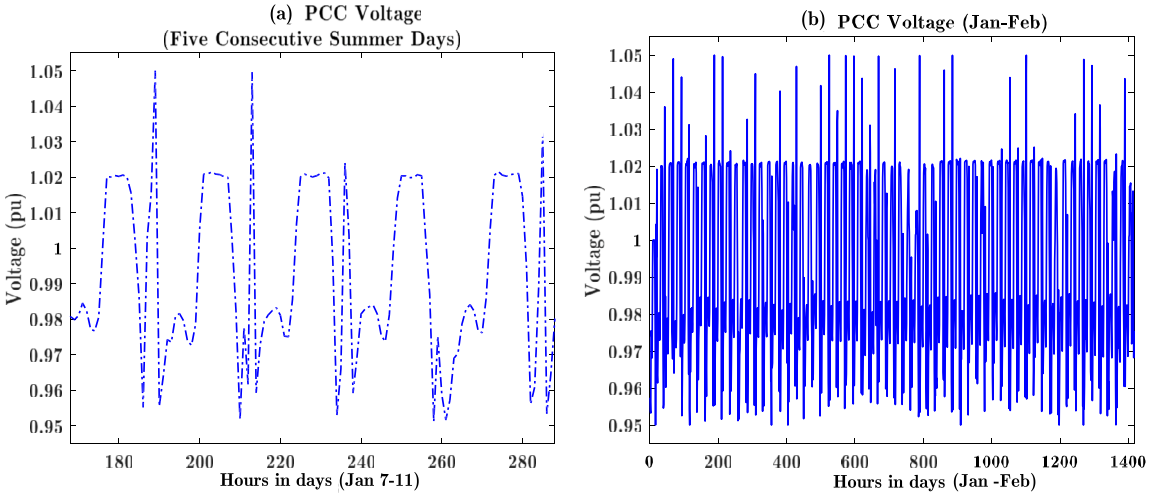


Fig. 7. PCC voltage during (a) five consecutive summer days, (b) summer months January - February with fixed rate electricity pricing scheme.

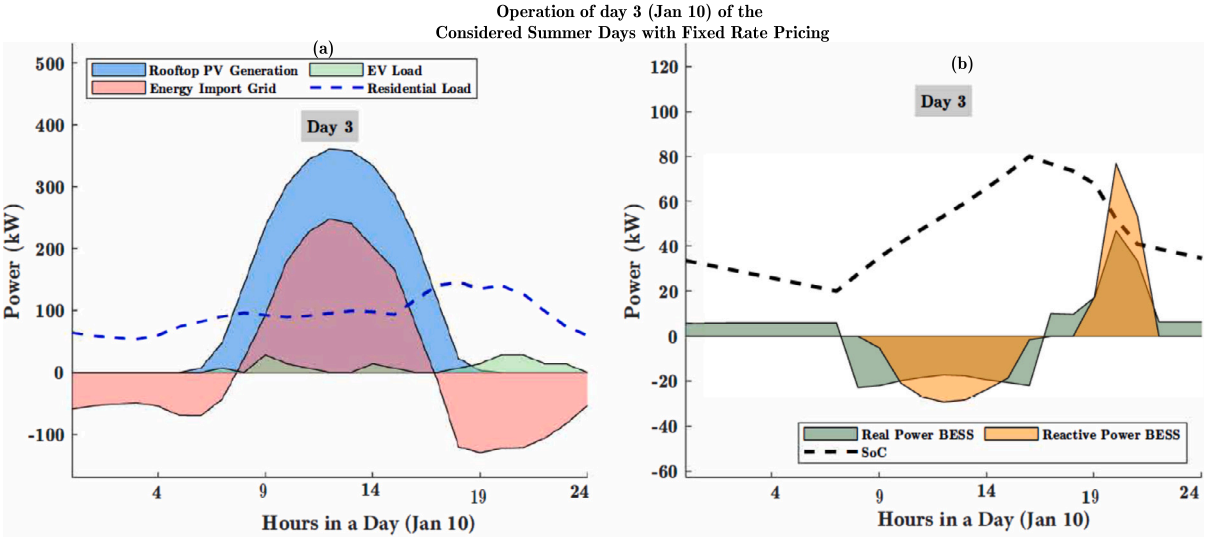


Fig. 8. (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS on day 3 of the selected summer days with fixed rate electricity pricing scheme.

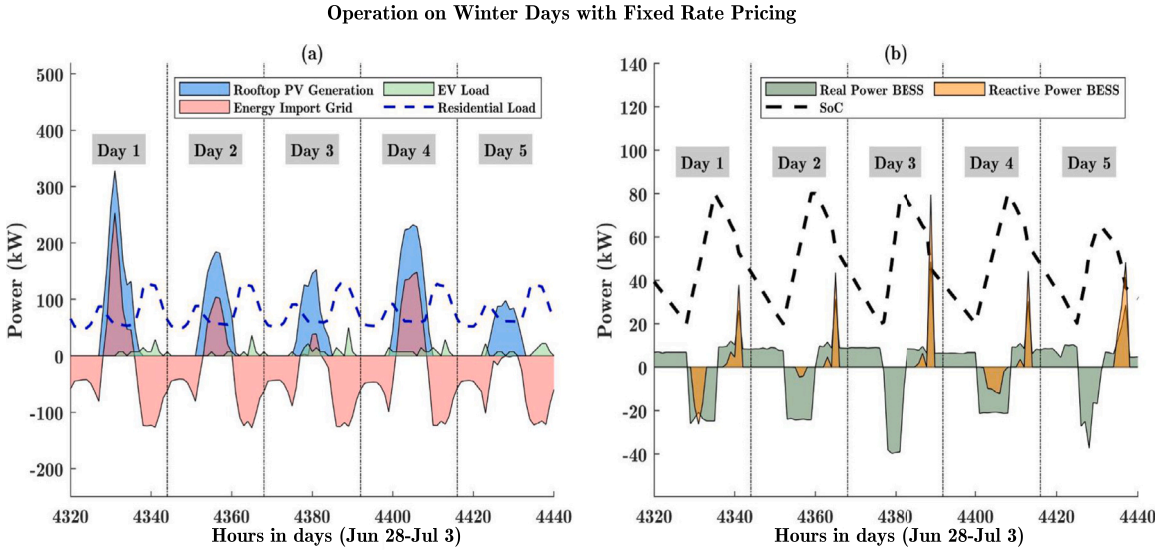


Fig. 9. Energy balance of the test residential complex for five consecutive winter days with fixed rate electricity pricing scheme. (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS.

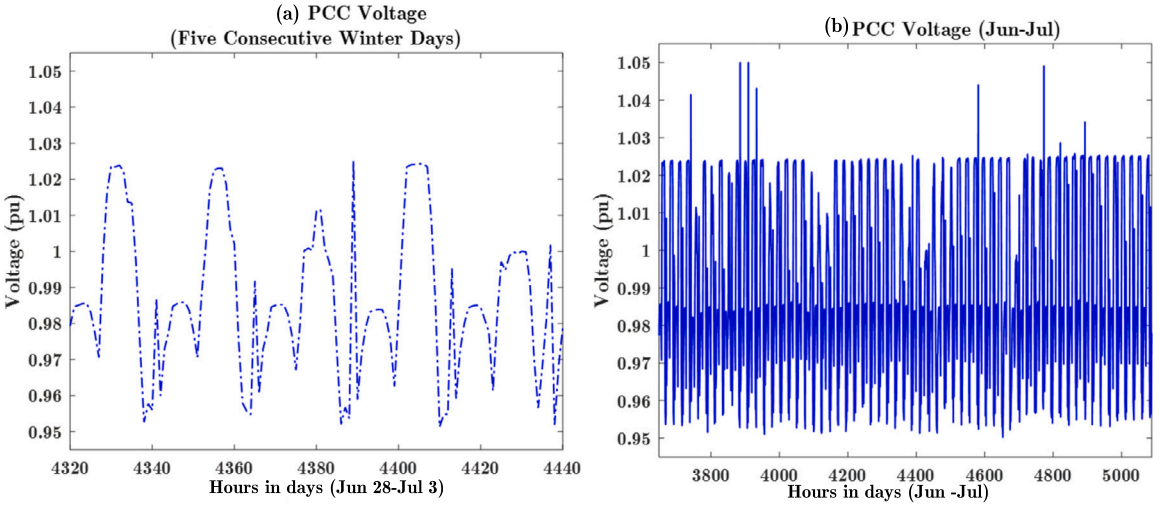


Fig. 10. PCC voltage during (a) five consecutive winter days, (b) winter months June - July with fixed rate electricity pricing scheme.

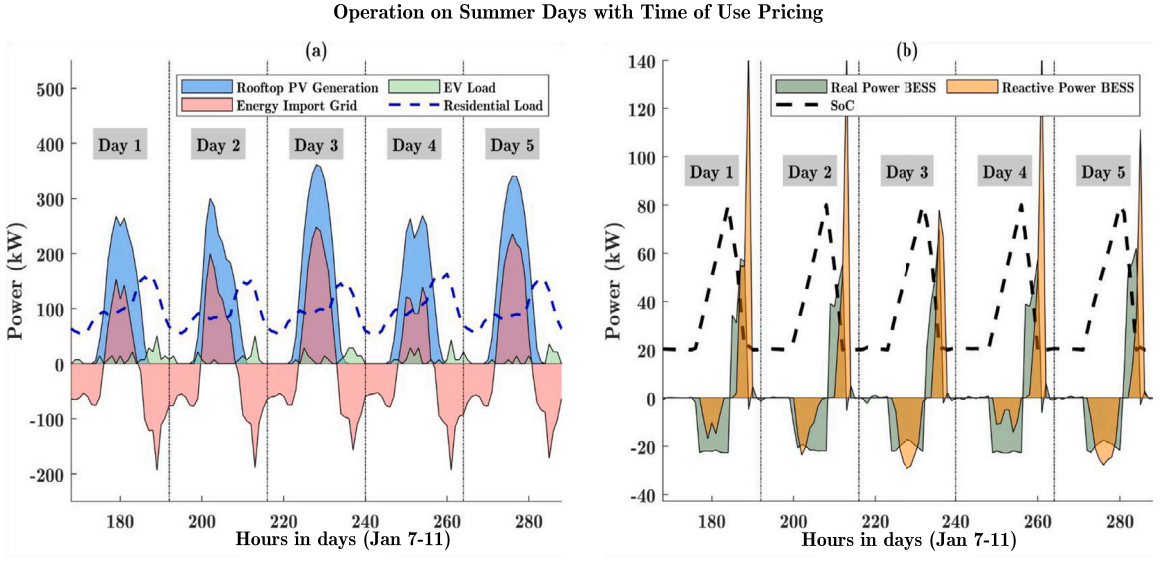


Fig. 11. Energy balance of the test residential complex for five consecutive summer days with ToU electricity pricing scheme. (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS.

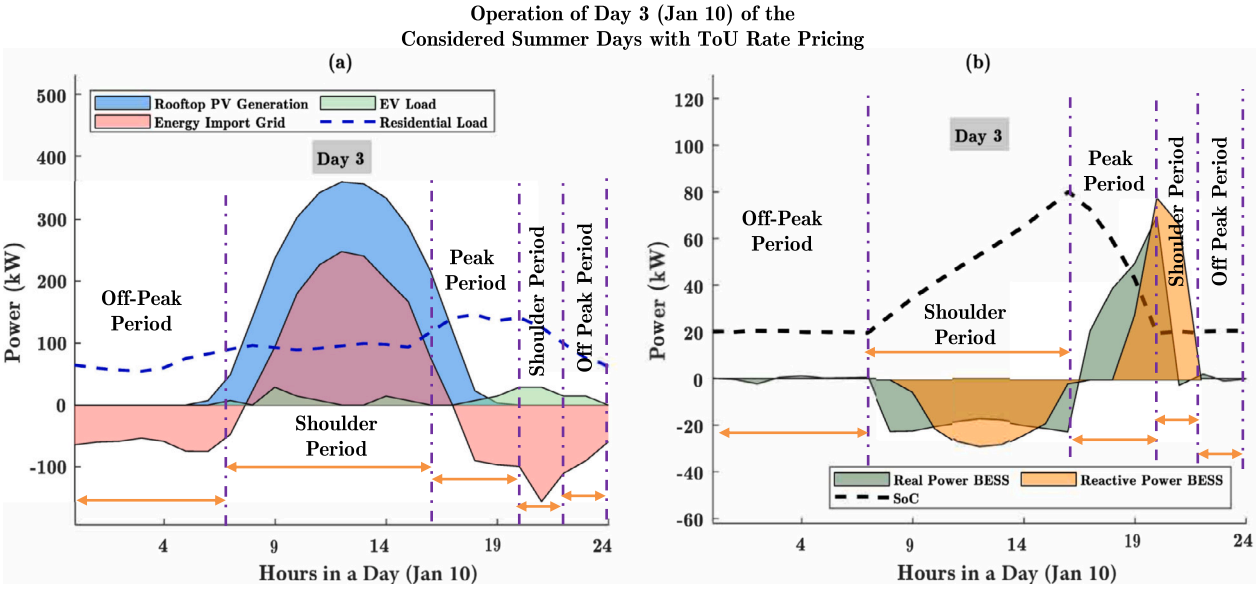


Fig. 12. (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS on day 3 of the selected summer days with TOU pricing scheme.

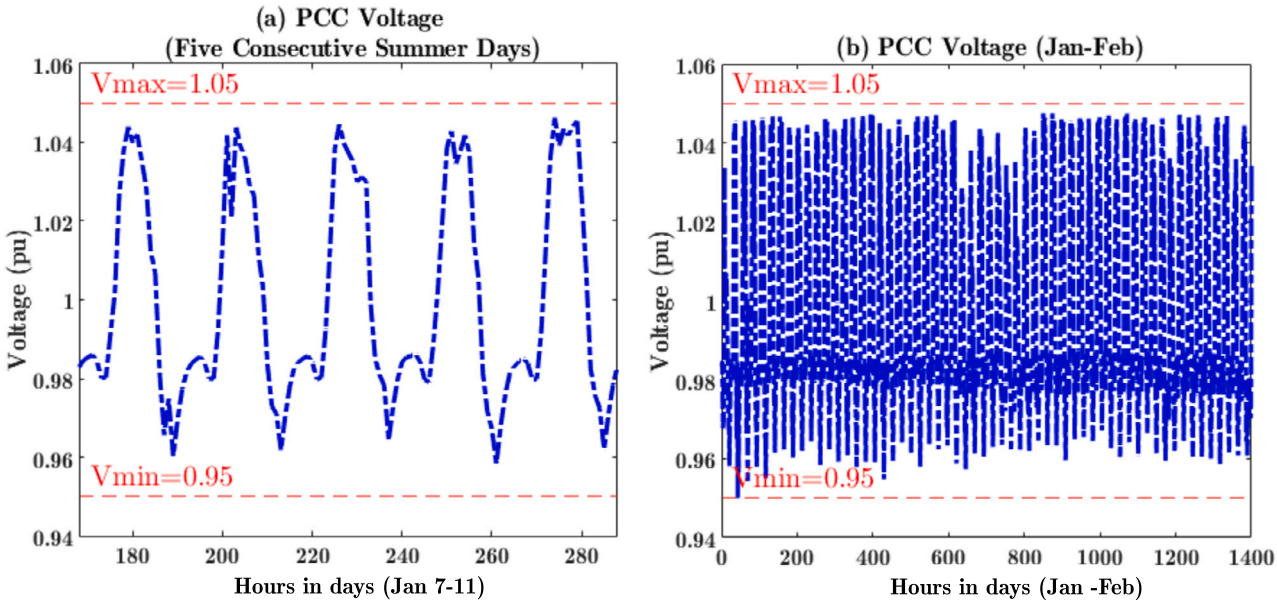


Fig. 13. PCC voltage during (a) five consecutive summer days, (b) summer months January - February with ToU pricing scheme.

is provided during peak solar periods and during peak demand periods to maintain PCC voltage within prescribed limits, mirroring the fixed pricing scenario. For a more comprehensive analysis, we illustrate the performance on the third day of selected summer days under the ToU pricing scheme, as depicted in Fig. 12. As in the fixed-rate scheme, the proposed EMS in the ToU pricing scheme ensures PCC voltage remains within 0.95-1.05 pu by using the BESS inverter for reactive power support. During peak solar hours, the BESS inverter absorbs reactive power to control voltage rise, while in the evening peak, it supplies reactive power to prevent voltage drops. Fig. 13 (a) and (b) illustrate PCC voltage over the considered five summer days and January-February.

Next, we evaluate the effectiveness of the proposed EMS for five consecutive winter days in Fig. 14. During winter, solar generation is relatively low, and the proposed methodology optimizes its operation under the ToU pricing scheme. To maximize economic benefits, the EMS employs a two-cycle strategy for the BESS. In the first cycle, the EMS ensures the BESS is fully charged at a controlled rate during the shoulder period by utilizing available solar energy. This stored energy is then used to supply the load demand in the residential complex, including residential and EV charging demands, during the peak period. The second BESS charging cycle begins at 11 pm when the shoulder period ends and the off-peak period starts. During this off-peak interval, the EMS purchases grid power to charge the BESS. This stored energy is discharged starting at 7 am the next day when the shoulder pricing period starts, emptying the BESS and preparing it for solar energy storage. This effectively reduces energy expenses, maximizes the utilization of BESS capacity, and takes advantage of the ToU pricing scheme. The figure also shows that the EMS utilizes reactive power support from the BESS inverter to maintain PCC voltage within the prescribed range, similar to the summer operation. For a more comprehensive analysis, we illustrate the operation on the fourth day of chosen winter days under the ToU pricing scheme, as depicted in Fig. 15.

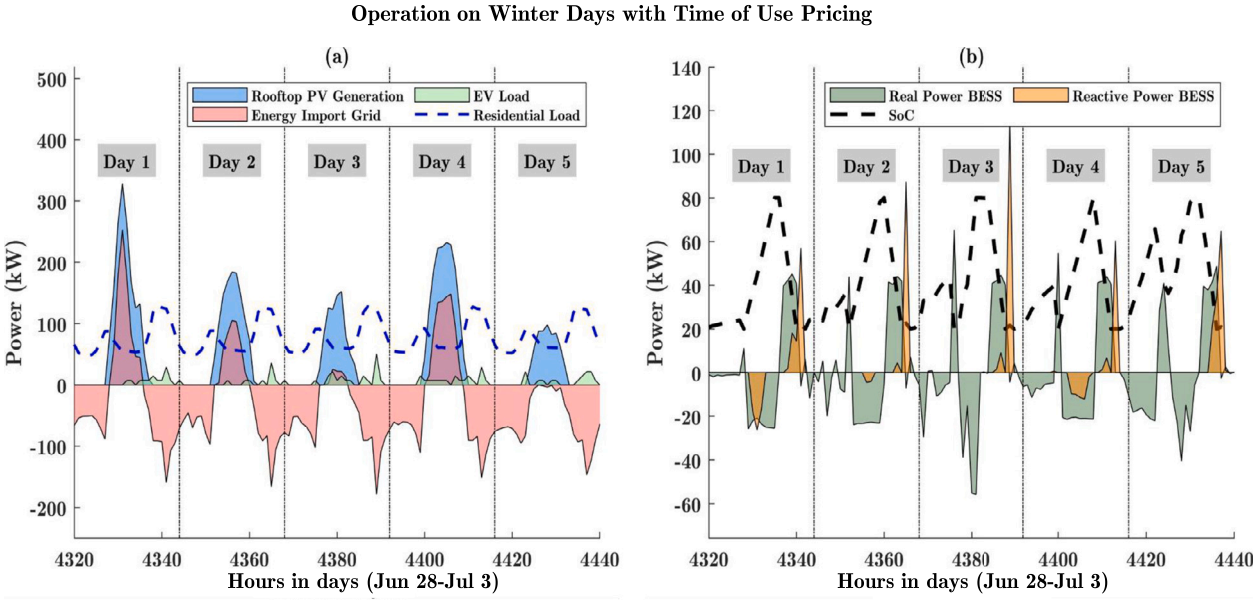


Fig. 14. Energy balance of the test residential complex for five consecutive winter days with ToU electricity pricing scheme. (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS.

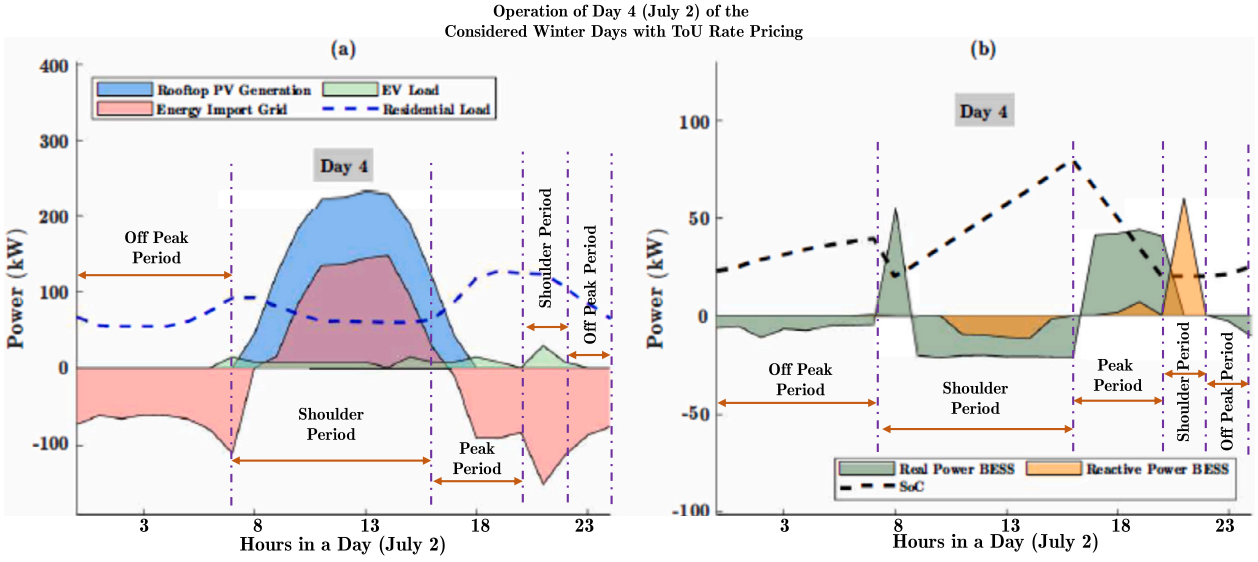


Fig. 15. (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS on day 3 of the selected winter days under TOU pricing scheme.

4.4. Comparison of proposed EMS with the existing approaches

This section presents a performance comparison of the proposed EMS with existing systems from the literature. As baseline approaches, we consider the EMSs presented in [54] and [31]. The EMS in [54] integrates a rule-based system with a Genetic Algorithm (GA) to optimize renewable energy and BESS operation. The rule-based system manages the charging and discharging of battery energy storage, ensuring renewable energy utilization and meeting basic system demands, while the GA refines power flows, maintains energy balance, and ensures battery SoC and power limits are respected, all while minimizing operational costs by reducing grid dependency. In contrast, the EMS in [31] employs a contribution factor-based strategy to adjust battery usage according to grid tariffs, aiming to minimize energy costs and reduce battery degradation in a grid-connected renewable energy system. In this comparative analysis, a one-year simulation study is conducted on the test system using the EMSs from [54] and [31]. Key metrics in this comparison include PCC voltage stability, battery degradation minimization, and overall economic performance optimization throughout the year.

First, we compare the operation of the proposed EMS with the EMSs from [54] and [31]. For this comparison, we refer to Fig. 11(a) and (b), Fig. 16(a) and (b), and Fig. 17(a) and (b), which show solar generation, load demand, grid power, BESS charging and discharging power, BESS SoC, and BESS reactive power support over five consecutive summer days in the test MG, using the proposed EMS, the EMS in [54], and the EMS in [31], respectively. As shown in Fig. 11(a) and (b), the proposed EMS not only regulates the charging and discharging of the BESS but also provides effective reactive power support to regulate PCC voltage. By considering load demand, solar generation, electricity price, and the intermittencies associated with both load and solar generation, the proposed EMS ensures that PCC voltage remains within prescribed limits through continuous reactive power adjustments. This is evident in Fig. 13(a) and (b), where the BESS reactive power support effectively stabilizes PCC voltage, a key advantage of the proposed EMS over the

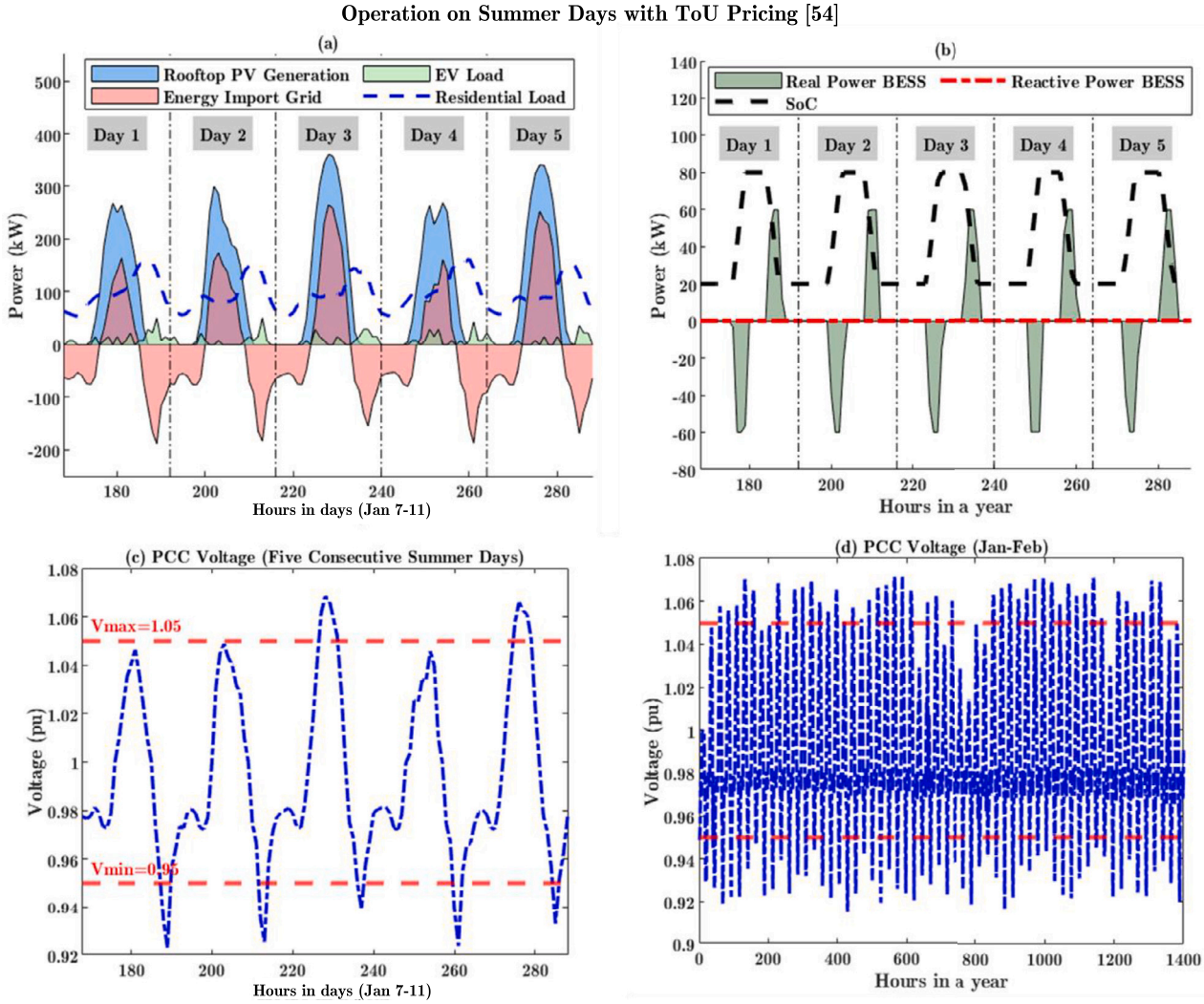


Fig. 16. Performance of the EMS presented in [54] under TOU pricing scheme: (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS of the selected summer days, PCC voltage during (c) the selected summer days, and (d) summer months January - February.

alternatives. A detailed discussion of this functionality has already been presented earlier. In contrast, the EMS in [54] focuses on optimizing operational costs by charging the BESS during solar hours and discharging it during periods of peak electricity prices, as shown in Fig. 16(a) and (b). However, this EMS does not provide reactive power support for PCC voltage regulation. As a result, while the BESS charges quickly and exports excess solar power to the grid, this leads to PCC voltage violations during high solar generation periods, as depicted in Fig. 16(c) and (d). Additionally, during evening peak hours, when electricity prices are high and BESS discharge partially meets the MG demand, significant grid imports further exacerbate PCC voltage violations due to the lack of reactive power support. Similarly, the EMS in [31] does not provide reactive power support from the BESS inverter to regulate PCC voltage, as shown in Fig. 17(a) and (b). Instead, it focuses on regulating power exchange between the MG and the utility grid by adjusting the BESS's charging and discharging cycles. However, without reactive power support, the rapid BESS charging and discharging during solar peaks and evening peak load periods lead to PCC voltage violations, particularly as the BESS is constrained by SOC limits and charging/discharging thresholds, as illustrated in Fig. 17(c) and (d). In summary, one of the key distinctions of the proposed EMS is its ability to provide reactive power support for PCC voltage regulation, a feature that is absent in both the EMS in [54] and the EMS in [31], resulting in better voltage stability and overall system performance.

Next, the BESS degradation performance of the proposed EMS is compared to the EMSs presented in [54] and [31]. As illustrated in Fig. 18, the proposed EMS achieves a significant reduction in BESS degradation, outperforming the baseline methods. Specifically, it reduces BESS degradation by 36% over one year of operation compared to the EMS in [54], which does not explicitly consider BESS degradation as an objective, highlighting the importance of integrating degradation minimization within the EMS framework. Furthermore, the proposed EMS demonstrates 23.6% less BESS degradation compared to the EMS in [31], despite the latter including degradation minimization as one of its goals. This difference is primarily due to the approach taken in regulating BESS operation. The EMS in [31] adjusts BESS charging and discharging to regulate power exchange between the MG and the utility grid in order to maintain PCC voltage. However, this results in more unregulated BESS cycling during periods of high solar generation and evening peak load, contributing to accelerated BESS degradation. In contrast, the proposed EMS optimizes BESS charging and discharging using a 24-hour ahead forecast of load and solar generation, while accounting for the associated intermittencies. Instead of relying on unregulated BESS cycling to regulate PCC voltage, the proposed EMS leverages reactive power support from the BESS, ensuring smoother and more controlled operation. This significantly reduces degradation and extends BESS lifespan, highlighting the superior performance of the proposed EMS compared to the baseline methods.

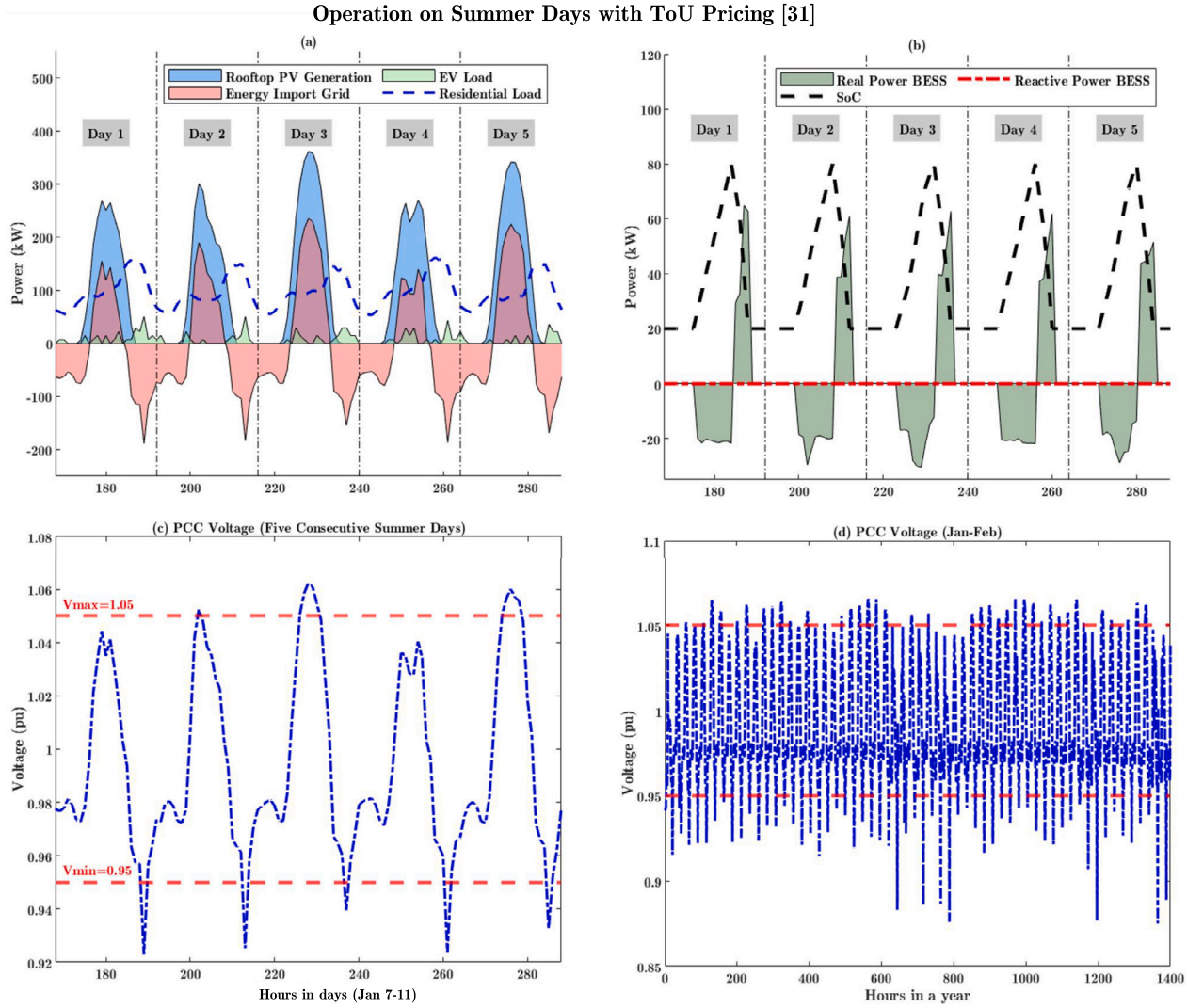


Fig. 17. Performance of the EMS presented in [31] under TOU pricing scheme: (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS of the selected summer days, PCC voltage during (c) the selected summer days, and (d) summer months January - February.

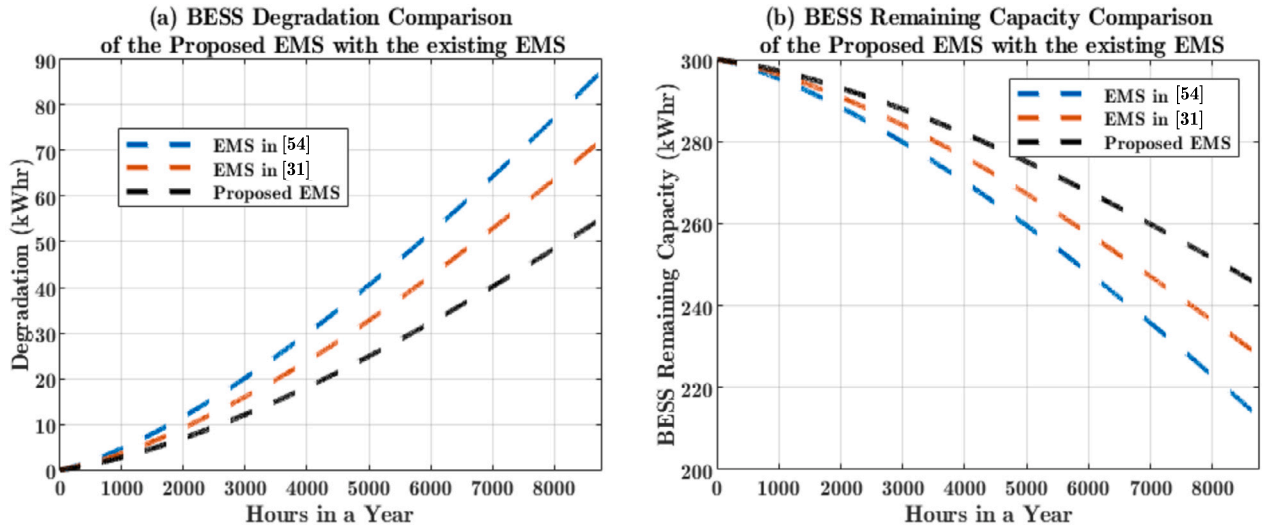


Fig. 18. BESS (a) Degradation, and (b) Remaining Capacity for One Year Operation of the MG Using Proposed EMS and EMS Presented in [54] and [31].

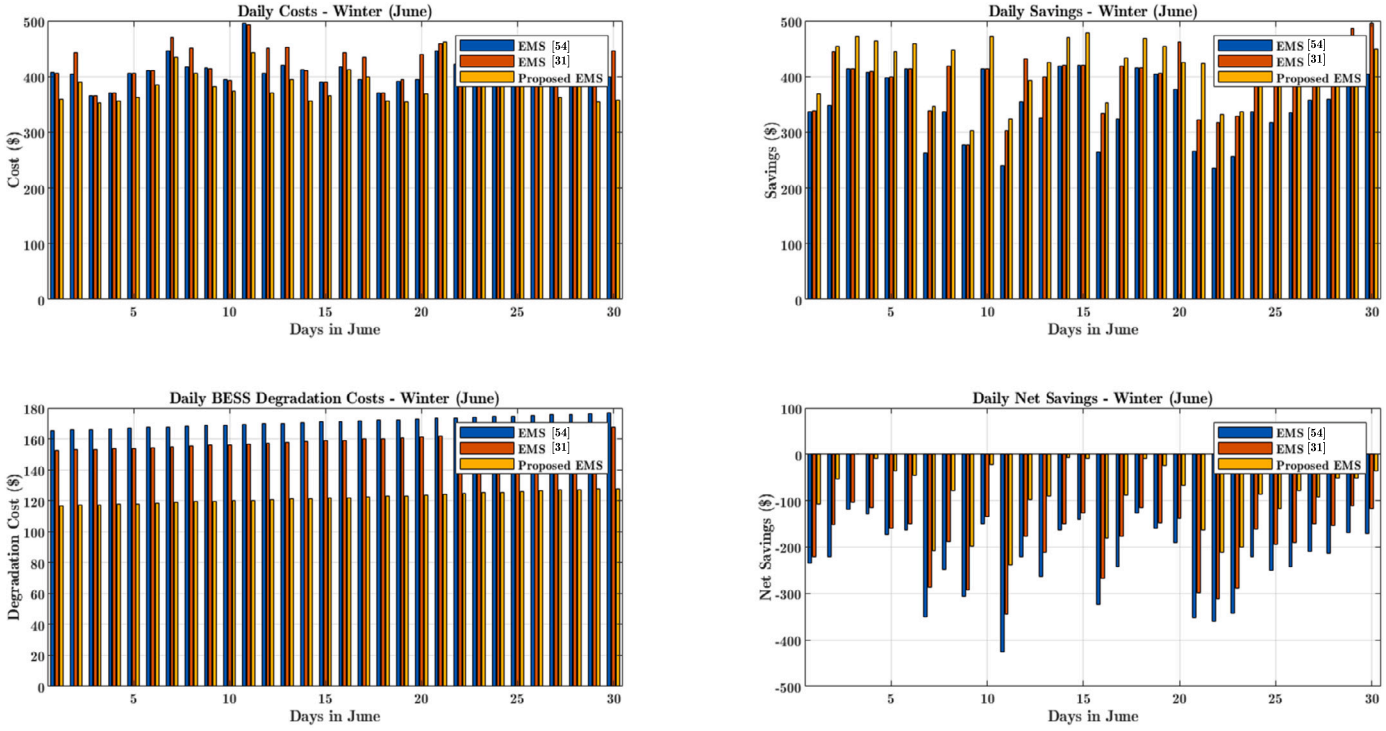


Fig. 19. Daily economic performance comparison of the proposed EMS with the EMS presented in [54] and [31] during the winter month June: (a) daily cost due to purchase of electricity from grid, (b) daily saving by utilizing harvested solar energy, (c) daily BESS degradation cost, and (d) daily net balance.

Lastly, we assess the economic performance of the proposed EMS compared to the EMSs from [54] and [31]. Figs. 19 and 20 depict the daily costs for grid electricity purchases and BESS degradation, calculated by multiplying the degraded BESS capacity by the per kWh cost of the BESS. These figures also show the savings from selling solar energy to the grid and using harvested solar and BESS energy to serve the load for January (summer) and June (winter). The net balance for the test system, using both the proposed EMS and the baseline approaches, is presented. The results demonstrate that the proposed EMS achieves better economic performance by optimizing BESS operation with a 24-hour prediction horizon, aligning with solar generation, load demand, and pricing. This is further confirmed by the year-long economic performance comparison shown in Fig. 21. Specifically, for this simulation study, EMS [54] has a yearly net balance of approximately -$46,000, EMS [31] has a yearly net balance of approximately -$38,000, while the proposed EMS achieves a net balance of approximately $10,000.

Overall, the proposed EMS outperforms the baseline approaches across all considered metrics PCC voltage stability, BESS degradation reduction, and economic performance by leveraging a 24-hour prediction horizon SMPC, reactive power management, and scenario-based optimization. These features establish it as a more efficient and reliable solution for MG energy management.

4.5. Validation of the proposed EMS through real-time simulation

To further evaluate the effectiveness of the proposed EMS and demonstrate its implementation in real-time scenarios, a real-time simulation of the test residential complex with the proposed EMS was conducted using the OPAL-RT real-time digital simulator platform. The setup involved two OP4512 simulators: OP4512 simulator 1 implemented the test residential complex, while OP4512 simulator 2 executed the proposed EMS. These simulators were interconnected through I/O boards. Both a schematic diagram and a photograph of this arrangement on the OPAL-RT platform are displayed in Fig. 22 (a) and (b). For a short overview on the OP4512 simulator, please refer to Appendix B.

The real-time operation of the EMS for day 3 of the summer period under the ToU pricing scheme was evaluated on the real-time simulation platform. Initial conditions for the BESS at the start of day 3, such as SOC and remaining capacity, were carried over from the prior MATLAB simulations. Fig. 23 (a) illustrates the behavior of the test residential complex during this specific day as captured by the real-time simulation. The results from this real-time simulation, including residential and EV charging load demands, solar generation, grid power exchange, and the BESS charging and discharging activities, along with its reactive power support, closely mirror those obtained from the MATLAB simulations shown in Fig. 8 (a) and (b). This congruence reaffirms the real-time viability and effectiveness of the proposed EMS in managing the BESS within the test residential complex, effectively replicating the performance results from the MATLAB simulations. Additionally, Fig. 23 (b) displays the real-time voltage profile at the PCC for the considered day, which remains within the prescribed limits. This is consistent with the voltage profile derived from the MATLAB simulations shown in Fig. 7. These results further validate the real-time applicability of the proposed EMS for efficient energy management in residential complexes and demonstrate its robustness in maintaining voltage stability and adapting to real-time conditions.

5. Conclusion

The proposed EMS for grid-connected residential complexes uses a stochastic model predictive control approach to optimize BESS operations over a 24-hour horizon, ensuring efficient use of harvested solar energy and minimizing grid dependence, resulting in cost savings and enhanced sustainability. The EMS integrates a battery degradation model to extend BESS lifespan and reduce replacement costs. It also provides reactive power support to maintain PCC voltage stability, ensuring safe and reliable operation. By generating multiple scenarios based on probability distributions of load demand, solar generation, and EV charging, the EMS effectively manages uncertainties, maintaining the reliability of the energy system under varying conditions. The integration of EV charging demand, crucial with the growing uptake of EVs, en-

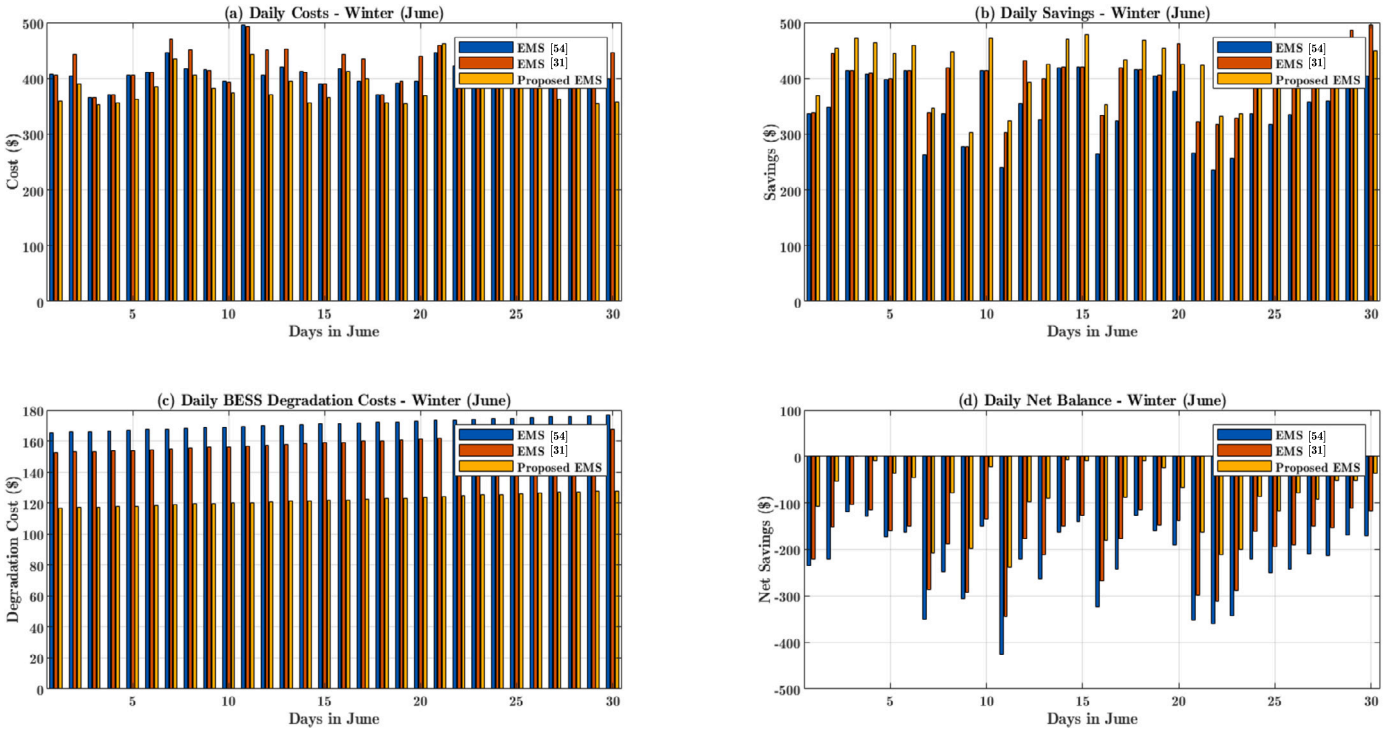


Fig. 20. Daily economic performance comparison of the proposed EMS with the EMS presented in [54] and [31] during the summer month January: (a) daily cost due to purchase of electricity from grid, (b) daily saving by utilizing harvested solar energy, (c) daily BESS degradation cost, and (d) daily net balance.

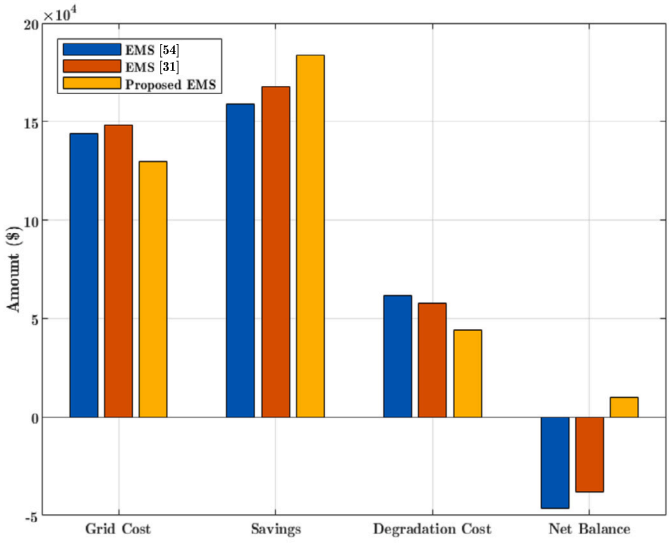


Fig. 21. Economic performance for one year operation of the test MG Using Proposed EMS and EMS Presented in [54] and [31].

ables seamless energy management despite added variability. The EMS adapts to both fixed-rate and time-of-use pricing schemes, optimizing energy costs by strategically charging and discharging the BESS according to electricity prices. Extensive simulations and real-time validations on the OPAL-RT platform confirm the EMS's effectiveness in managing diverse conditions, including variable solar generation and fluctuating load demands. The SMPC-based approach maximizes economic benefits while ensuring operational efficiency. The robustness of the EMS through scenario-based planning emphasizes its practicality for deployment in real-world residential complexes. A comparative performance analysis between the proposed EMS and existing approaches demonstrates the superior ability of the EMS to maintain PCC voltage stability, reduce operational and battery degradation costs, and manage uncertainties.

While the proposed EMS provides significant contributions to sustainable energy management, certain limitations remain, which present opportunities for future work:

- *Computational Complexity:* Managing multiple scenarios simultaneously increase the computational load. Future work could explore scenario reduction techniques or simplified models to improve scalability for larger systems.
- *EV User Behavior Modeling:* The current study considers the first-come, first-serve strategy but does not incorporate detailed behavioral aspects of EV users, such as willingness to wait, reservation systems, or responses to dynamic pricing. Future work could develop behavior-specific models based on the characteristics and preferences of residents within different types of built environments.
- *Generality of the EMS:* While the proposed EMS is designed to be adaptable, future work could further generalize it to suit a variety of built environments, such as residential complexes, commercial buildings, and industrial precincts. Additionally, the EMS could be extended to operate across multiple geographically separated buildings that are owned by the same organization, enabling them to function as a unified energy management system.
- *VPP and Market Integration:* Integrating the EMS into a virtual power plant framework could enable coordinated control with other distributed energy resources and participation in ancillary services, enhancing both economic and operational benefits.
- *Infrastructure Expansion and V2G:* Incorporating vehicle-to-grid capabilities and expanding infrastructure could provide additional value by supporting grid stability and flexibility.

CRediT authorship contribution statement

M.I. Saleem: Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **S. Saha:** Writing – original draft, Val-

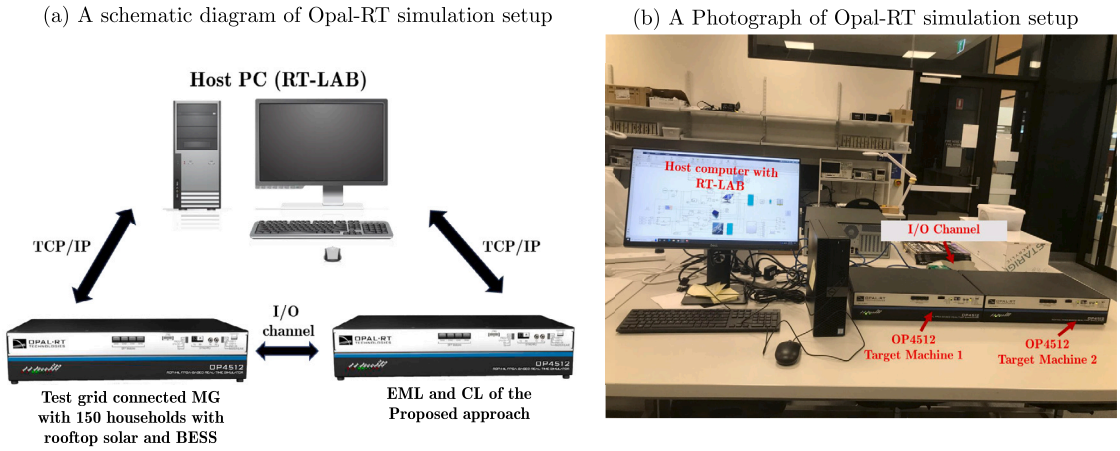


Fig. 22. (a) A schematic diagram showcasing the setup for real-time simulation using the OPAL-RT real-time digital simulator platform. (b) A photograph of the real-time simulation set-up.

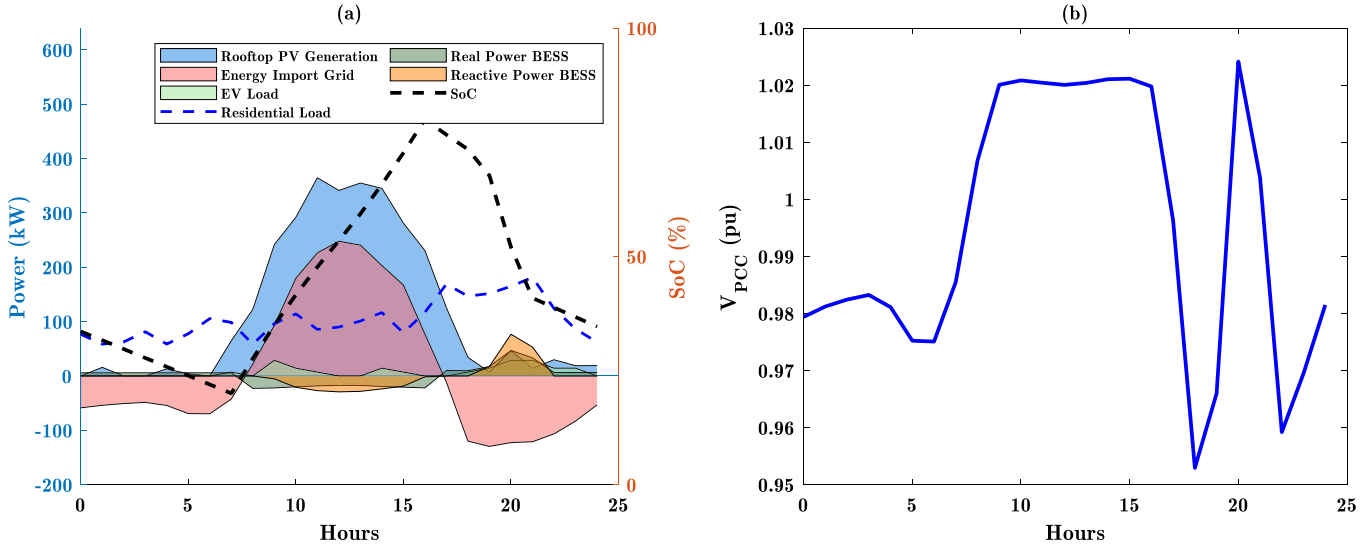


Fig. 23. Real time simulation result of the test RC operation on Day 3 of the selected summer days shown in Fig. 11 under the ToU pricing scheme: (a) Load demand, solar generation, and grid power, (b) BESS SOC, real and reactive power output of BESS.

idation, Methodology, Conceptualization. **U. Izhar:** Writing – review & editing. **L. Ang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Generation of probable 24-hour ahead scenarios for SMPC optimization in proposed EMS

The proposed EMS utilizes a set of N probable scenarios to handle uncertainties in solar PV generation, residential load demand, and EV charging over a 24-hour prediction horizon. Each scenario reflects a particular realization of future operating conditions, helping the proposed EMS optimize BESS operation effectively. In the following, we describe the procedure to generate these scenarios.

A.1. Step 1: generate uncorrelated random samples for each hour in the 24-hour horizon

At the hour h_i, $K = 1000$ uncorrelated random samples for residential load demand, solar generation, and EV charging demand are generated for each of the subsequent 24 hours (i.e., from h_{i+1} to h_{i+24}). This value of K is chosen to sufficiently capture the variability in the operating conditions while maintaining computational efficiency. These random samples are drawn independently from their respective probability distributions defined earlier in equations (6), (7), and (16).

The uncorrelated random draws for each hour are represented as follows:

$$\mathcal{X}_{\text{Load}}^{h_{i+p}} = \left\{ L_1^{h_{i+p}}, L_2^{h_{i+p}}, \ldots, L_K^{h_{i+p}} \right\}, \quad p = 1, \ldots, 24 \tag{53}$$

$$\mathcal{X}_{\text{Solar}}^{h_{i+p}} = \left\{ S_1^{h_{i+p}}, S_2^{h_{i+p}}, \ldots, S_K^{h_{i+p}} \right\}, \quad p = 1, \ldots, 24 \tag{54}$$

$$\mathcal{X}_{\text{EV}}^{h_{i+p}} = \left\{ E_1^{h_{i+p}}, E_2^{h_{i+p}}, \ldots, E_K^{h_{i+p}} \right\}, \quad p = 1, \ldots, 24 \tag{55}$$

These random samples are independently generated from their respective normal distributions. The value of K is selected to balance between capturing enough variability in the random samples and ensuring computational efficiency. Typically, a higher K provides a better representation of uncertainty, but it also increases computational complexity. In this study, K is chosen based on trial simulations to ensure accuracy without significantly impacting the computation time.

A.2. Step 2: generate multivariate correlation matrix

In this step, we establish correlations between the random samples of load demand, solar generation, and EV charging, both across different hours (self-correlation) and between different variables (cross-correlation).

A.2.1. Self-correlation matrix

For each variable load demand, solar generation, and EV charging we model the temporal self-correlation across the 24-hour horizon to account for the dependency of each variable on its previous and future values. The self-correlation matrix for load demand, denoted as $\Sigma_{\text{Load}} \in \mathcal{R}^{24\times24}$ where each element represents the correlation between the load at different hours over the 24-hour horizon. This matrix captures the temporal dependencies between consecutive and non-consecutive hours as given by:

$$\Sigma_{\text{Load}} = \begin{bmatrix} 1 & \Sigma_{\text{Load}}^{|h_{i+1}-h_{i+2}|} & \Sigma_{\text{Load}}^{|h_{i+1}-h_{i+3}|} & \cdots & \Sigma_{\text{Load}}^{|h_{i+1}-h_{i+24}|} \\ \Sigma_{\text{Load}}^{|h_{i+2}-h_{i+1}|} & 1 & \Sigma_{\text{Load}}^{|h_{i+2}-h_{i+3}|} & \cdots & \Sigma_{\text{Load}}^{|h_{i+2}-h_{i+24}|} \\ \Sigma_{\text{Load}}^{|h_{i+3}-h_{i+1}|} & \Sigma_{\text{Load}}^{|h_{i+3}-h_{i+2}|} & 1 & \cdots & \Sigma_{\text{Load}}^{|h_{i+3}-h_{i+24}|} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma_{\text{Load}}^{|h_{i+24}-h_{i+1}|} & \Sigma_{\text{Load}}^{|h_{i+24}-h_{i+2}|} & \Sigma_{\text{Load}}^{|h_{i+24}-h_{i+3}|} & \cdots & 1 \end{bmatrix} \tag{56}$$

The self-correlation term for load demand at hours h_{i+a} and h_{i+b}, where $a, b \in \{1, \ldots, 24\}$ is given by:

$$\Sigma_{\text{Load}}^{|h_{i+a}-h_{i+b}|} = \frac{\text{Cov}(\mathcal{X}_{\text{Load}}^{h_{i+a}}, \mathcal{X}_{\text{Load}}^{h_{i+b}})}{\sigma_{\overline{\mathbf{P}}_{\text{D,H}}^{h_{i+a}}} \cdot \sigma_{\overline{\mathbf{P}}_{\text{D,H}}^{h_{i+b}}}} \tag{57}$$

where

$$\text{Cov}(\mathcal{X}_{\text{Load}}^{h_{i+a}}, \mathcal{X}_{\text{Load}}^{h_{i+b}}) = \frac{1}{K}\sum_{d=1}^{K}\left(L_d^{h_{i+a}} - \mu_{\overline{\mathbf{P}}_{\text{D,H}}}^{h_{i+a}}\right)\left(L_d^{h_{i+b}} - \mu_{\overline{\mathbf{P}}_{\text{D,H}}}^{h_{i+b}}\right)$$

The terms $\mu_{\overline{\mathbf{P}}_{\text{D,H}}}^{h_{i+a}}$ and $\sigma_{\overline{\mathbf{P}}_{\text{D,H}}}^{h_{i+a}}$ are the mean and the standard deviation of the load demand at hour h_{i+a}, as presented in (7). Similarly self-correlations for 24 hour horizon solar generation and EV charging demand, Σ_{Solar} and Σ_{EV} can be determined.

A.2.2. Cross-correlation matrix

Next, we determine the cross-correlation matrix between different variables. The cross-correlation matrix $\Sigma_{\text{Load-Solar}} \in \mathcal{R}^{24\times24}$ represents the correlation between load demand and solar generation over the 24-hour horizon. Each element in this matrix represents the correlation between load demand at hour h_{i+a} and solar generation at hour h_{i+b}, where $a, b \in \{1, \ldots, 24\}$. The matrix is given by:

$$\Sigma_{\text{Load-Solar}} = \begin{bmatrix} \Sigma_{\text{Load-Solar}}^{|h_{i+1}-h_{i+1}|} & \Sigma_{\text{Load-Solar}}^{|h_{i+1}-h_{i+2}|} & \Sigma_{\text{Load-Solar}}^{|h_{i+1}-h_{i+3}|} & \cdots & \Sigma_{\text{Load-Solar}}^{|h_{i+1}-h_{i+24}|} \\ \Sigma_{\text{Load-Solar}}^{|h_{i+2}-h_{i+1}|} & \Sigma_{\text{Load-Solar}}^{|h_{i+2}-h_{i+2}|} & \Sigma_{\text{Load-Solar}}^{|h_{i+2}-h_{i+3}|} & \cdots & \Sigma_{\text{Load-Solar}}^{|h_{i+2}-h_{i+24}|} \\ \Sigma_{\text{Load-Solar}}^{|h_{i+3}-h_{i+1}|} & \Sigma_{\text{Load-Solar}}^{|h_{i+3}-h_{i+2}|} & \Sigma_{\text{Load-Solar}}^{|h_{i+3}-h_{i+3}|} & \cdots & \Sigma_{\text{Load-Solar}}^{|h_{i+3}-h_{i+24}|} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma_{\text{Load-Solar}}^{|h_{i+24}-h_{i+1}|} & \Sigma_{\text{Load-Solar}}^{|h_{i+24}-h_{i+2}|} & \Sigma_{\text{Load-Solar}}^{|h_{i+24}-h_{i+3}|} & \cdots & \Sigma_{\text{Load-Solar}}^{|h_{i+24}-h_{i+24}|} \end{bmatrix} \tag{58}$$

Each element $\Sigma_{\text{Load-Solar}}^{|h_{i+a}-h_{i+b}|}$ is calculated as the correlation between the load demand at hour h_{i+a} and the solar generation at hour h_{i+b}, using the following equation:

$$\Sigma_{\text{Load-Solar}}^{|h_{i+a}-h_{i+b}|} = \frac{\text{Cov}(\mathcal{X}_{\text{Load}}^{h_{i+a}}, \mathcal{X}_{\text{Solar}}^{h_{i+b}})}{\sigma_{\overline{\mathbf{P}}_{\text{D,H}}^{h_{i+a}}} \cdot \sigma_{\overline{\mathbf{P}}_{\text{PV}}^{h_{i+b}}}} \tag{59}$$

where the covariance between load demand and solar generation is defined as:

$$\text{Cov}(\mathcal{X}_{\text{Load}}^{h_{i+a}}, \mathcal{X}_{\text{Solar}}^{h_{i+b}}) = \frac{1}{K}\sum_{d=1}^{K}\left(L_d^{h_{i+a}} - \mu_{\overline{\mathbf{P}}_{\text{D,H}}}^{h_{i+a}}\right)\left(S_d^{h_{i+b}} - \mu_{\overline{\mathbf{P}}_{\text{PV}}}^{h_{i+b}}\right) \tag{60}$$

The cross-correlation matrix $\Sigma_{\text{Load-Solar}}$ captures the relationship between load demand and solar generation across the 24-hour horizon, reflecting the interactions between these variables. Similarly, the cross-correlation matrices $\Sigma_{\text{Load-EV}} \in \mathcal{R}^{24\times24}$ and $\Sigma_{\text{Solar-EV}} \in \mathcal{R}^{24\times24}$ are computed to represent the relationships between load demand and EV charging, and solar generation and EV charging, respectively.

A.2.3. Complete correlation matrix

In this step, we derive the complete multivariate correlation matrix $\Sigma_{\text{Total}} \in \mathcal{R}^{72\times72}$, which incorporates both the self-correlation of each variable and the cross-correlation between load demand, solar generation, and EV charging across the 24-hour horizon as shown below:

$$\Sigma_{\text{Total}} = \begin{bmatrix} \Sigma_{\text{Load}} & \Sigma_{\text{Load-Solar}} & \Sigma_{\text{Load-EV}} \\ \Sigma_{\text{Solar-Load}} & \Sigma_{\text{Solar}} & \Sigma_{\text{Solar-EV}} \\ \Sigma_{\text{EV-Load}} & \Sigma_{\text{EV-Solar}} & \Sigma_{\text{EV}} \end{bmatrix} \tag{61}$$

A.3. Step 3: scenario generation using the correlation matrix

In this step, N number of 24-hour ahead scenarios are generated using the derived correlation matrix Σ_{Total}. For each scenarios presented in section 3.1, following steps are undertaken:

- *Uncorrelated Random Sampling*: At each hour h_i, we generate uncorrelated random samples for load demand, solar generation, and EV charging demand, independently drawn from their respective probability distributions as described in Step 1 (53)-(55). The generated random samples for each hour across the 24-hour horizon are represented as:

$$\mathcal{X}_n \in \mathcal{R}^{72\times1} = \left[\mathcal{X}_{\text{Load,n}}^{h_{i+1}}, \cdots, \mathcal{X}_{\text{Load,n}}^{h_{i+24}}, \mathcal{X}_{\text{Solar,n}}^{h_{i+1}}, \cdots, \mathcal{X}_{\text{Solar,n}}^{h_{i+24}}, \mathcal{X}_{\text{EV,n}}^{h_{i+1}} \cdots \mathcal{X}_{\text{EV,n}}^{h_{i+24}}\right]^T \tag{62}$$

- *Cholesky Decomposition*: Next applying the Cholesky Decomposition on the derived multivariate correlation matrix Σ_{Total} (61), a lower triangular matrix $\Lambda \in \mathcal{R}^{72\times72}$ is yielded:

$$\Sigma_{\text{Total}} = \Lambda\Lambda^{\top} \tag{63}$$

This is followed by multiplying the uncorrelated random samples $\mathcal{X}$ by Λ to determine the correlated random samples for load demand, solar, and EV charging demand:

$$\mathcal{Y}_n = \Lambda\mathcal{X}_n \tag{64}$$

- *Scaling, Normalization and Scenario Construction*: After determining the correlated random samples $\mathcal{Y}_n$ of load demand, solar, and EV charging demand for next 24 hours $h_{i+1}, \cdots, h_{i+24}$, they are scaled and normalized based on the means and standard deviations of these variables of corresponding hours. Then these normalized correlated random samples are rearranged to determine a 24 hour ahead operating scenario of the MG as in (35).

These steps are repeated for each of the N scenarios to generate probable 24-hour ahead operating conditions of the MG. The total number of scenarios N is chosen by balancing the trade-off between computational complexity and simulation accuracy. A larger N provides better representation of possible future operating conditions, but it also increases the computational burden of solving the optimization problem. In this study, we used $N = 1000$ scenarios to adequately capture the range of

potential outcomes while maintaining a computationally feasible optimization process for the proposed EMS. These 1000 probable 24-hour ahead scenarios for each hour h_i can be generated off-line and stored as a scenario bank. The scenario bank can be accessed to formulate the optimization problem at the h_i hour, as outlined in Step 2 of the proposed EMS.

Appendix B. Description of OP4512

The OP4512 is a high-performance real-time simulator designed for power system studies, offering precise modeling of electrical grids, microgrids, and renewable energy integration. It features multi-core processors for fast simulations, supporting complex systems with high accuracy. The simulator's hardware-in-the-loop capability allows seamless interaction with real-world devices, making it ideal for validating controllers and equipment. Its flexible I/O options provide integration with external hardware like relays and sensors. Additionally, the OP4512 enables real-time data acquisition and control, allowing researchers to monitor, adjust, and optimize system performance during dynamic simulations of power system events.

Data availability

The data supporting the findings of this study, including yearly load demand and rooftop solar generation on an hourly basis, are available at the following link: https://drive.google.com/drive/folders/1XMkrhZrh-8I15SrLw9JxULVLej0HUUQj?usp=sharing. These datasets were utilized to derive the probability distributions for hourly residential load demand and rooftop solar generation, as detailed in Sections 2.1 and 2.2.1 for the case study. The EV charging demand for the case study was generated using the model presented in Section 2.2.2, and the corresponding hourly EV charging demand data used in the case study is also included in the shared link. By following the EMS steps outlined in Section 3 and the Appendix, readers will be able to reproduce the EMS framework and validate the results presented in this paper.

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